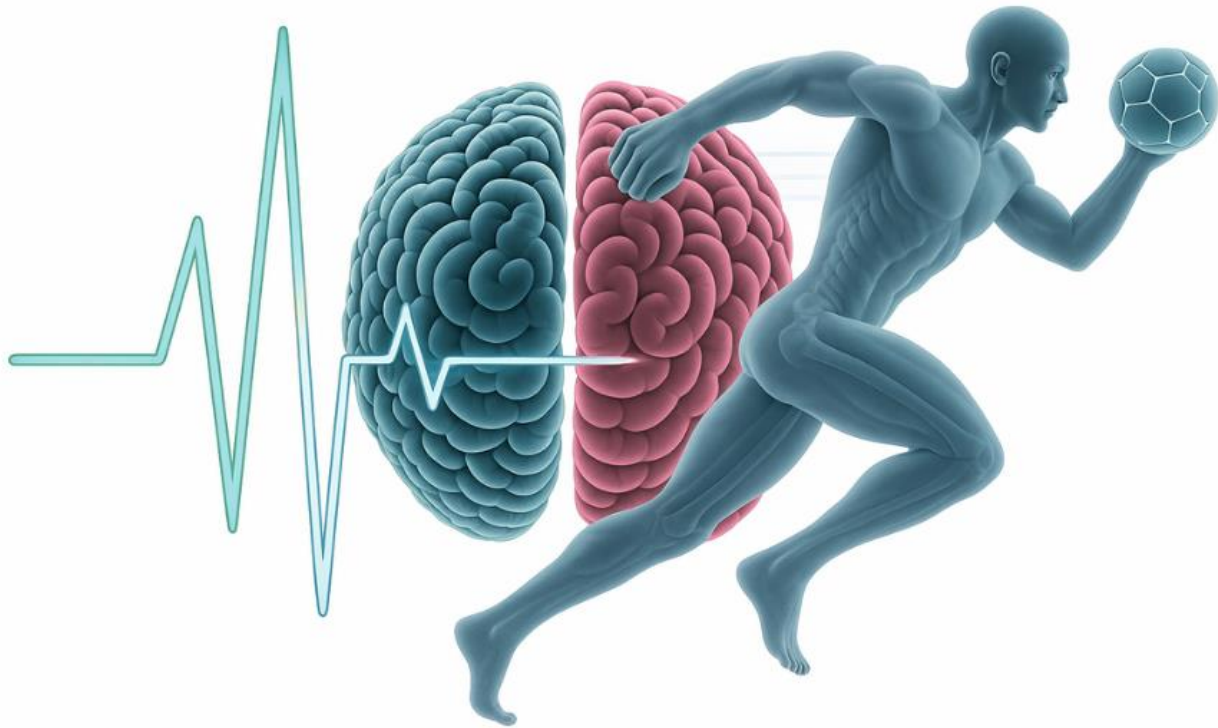


Book of Abstracts

ISBN: 978-65-5973-545-7

4 to 6 of December 2025
Porto Alegre, RS, Brazil



III SNNSEM

**III SYMPOSIUM ON NEUROPHYSIOLOGY AND
NEUROMODULATION IN SPORT, EXERCISE
AND MOTOR REHABILITATION**

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Catálogo na Publicação (CIP)

III Symposium on neurophysiology and neuromodulation in Sport, exercise and motor rehabilitation. / organização de Cássio Víctora Ruas, Tércio Apolinário de Souza, Ronei Silveira Pinto. Porto Alegre, RS : III SNNSEM, 2025.

PDF. : il.

ISBN 978-65-5973-545-7

1. neurophysiology . 2. exercise 3. rehabilitation. 4. Neuromodulation
I. Cássio Víctora Ruas II. Tércio Apolinário de Souza
III. Ronei Silveira Pinto.

CDU 796:615.8

BIBLIOTECA DA ESCOLA DE EDUCAÇÃO FÍSICA, FISIOTERAPIA E DANÇA.

Message from the Presidents of the III SNNSEM

Dear speakers, professionals and students who participated in the III Symposium on Neurophysiology and Neuromodulation in Sport, Exercise, and Motor Rehabilitation (III SNNSEM)

Thank you very much for your presence and for your excellent participation and contributions, which were fundamental to the success of this year's event!

The aim of this edition was to present and discuss topics across three main areas: *Exercise Neurophysiology and Neuromodulation; Neuromuscular Adaptations to Exercise; and Sport, Motor Behavior and Rehabilitation*. In particular, the Symposium provided an opportunity to further expand the knowledge of current neurotechnologies used to obtaining neuromuscular and neurophysiological measurements within the fields of sport and health, and their clinical applications in the motor rehabilitation of patients with neurological disorders.

The event featured a series of lectures delivered by world-leading national and international researchers in the fields of exercise science, motor behavior, neurophysiology and neurorehabilitation. In addition, a total of 43 abstracts were submitted by students and professionals and presented in poster format. In this book of abstracts, we present all of the high-quality abstracts showcased during the Symposium.

The SNNSEM was first conceived by Profs. Cassio, Ronei and Tercio in 2023, and it is an honor for us to have reached its third annual edition. We greatly appreciate the participation of all attendees in the event, whose engagement and discussions contributed to exercise neurophysiology research advancement.

We hope to see you again soon for the next edition of the event!

Sincerely,

Cássio Víctora Ruas
Tércio Apolinário de Souza
Ronei Silveira Pinto
Presidents of the SNNSEM



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Cristiani Junqueira (UFMG)



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Edith Cowan University, Australia



Anthony Blazeovich

Edith Cowan University, Australia



Marco Vaz

Universidade Federal do Rio Grande
do Sul



Cássio Ruas

Universidade Estadual de Campinas



Camila Lima

Universidade Estadual de Campinas



Rita de Cássia Silveira

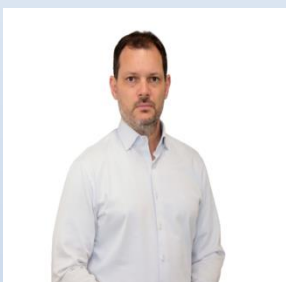
Universidade Federal do Rio Grande
do Sul



Gabriela Castellano
Universidade Estadual de Campinas



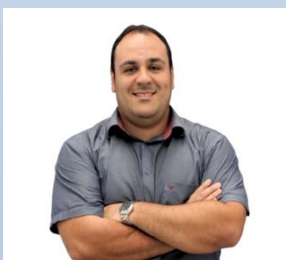
Leonardo Tartaruga
Università di Pavia, Italia



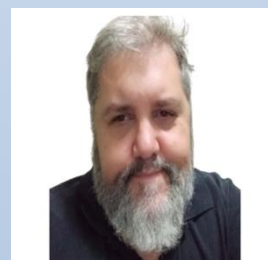
Alexandre Brandão
Pontifícia Universidade Católica
de Campinas



Sandro Freitas
Universidade de Lisboa



Maicon Albuquerque
Universidade Federal de
Minas Gerais



Guilherme Lage
Universidade Federal de
Minas Gerais



Luis Peñailillo
Universidad Andres Bello,
Chile



Carlos Cruz-Montecinos
Universidad de Chile,
Chile

PROGRAM

Day 1



III SNNSEM
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04/12
(Thursday)

09:00 - 09:30

Opening Cerimony

09:30 - 11:00

Opening Lecture: Ken Nosaka

Neurophysiological characteristics of eccentric exercise in relation to sports, health and rehabilitation

11:30 - 12:30

Leonardo Tartaruga

Behavioural energetics in human locomotion: mechanical determinants of energy cost

14:30 - 15:30

Rita de Cássia Silveira

Stimulating the premature brain: early impacts on neurodevelopmental outcome

16:30 - 17:30

Gabriela Castellano

The Origins of the EEG Signal and Approaches to Its Processing and Analysis

17:30 - 18:30

Sandro Freitas

Unveiling a human motor power approach as (probably) the most feasible biomarker of healthy aging



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04/12/25 -
06/12/25

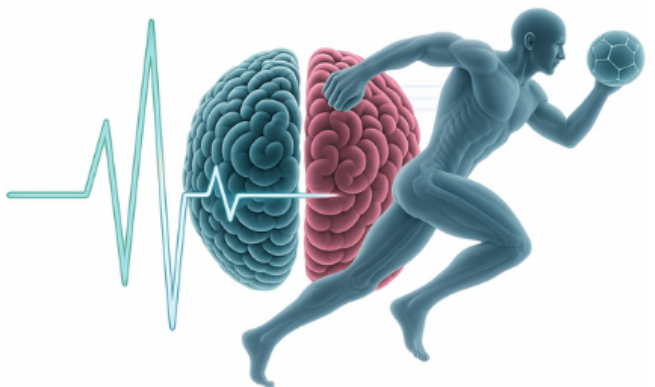


In-Person
Event



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Day 2



III SNNSEM

III SYMPOSIUM ON NEUROPHYSIOLOGY AND
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04/12/25 a
06/12/25



Evento
Presencial



Instituto de Ciências Básicas
da Saúde (ICBS) - UFGRS

PROGRAM

05/12
(Friday)

09:30 - 11:00

Marco Vaz

Neuromuscular Electrical Stimulation for
Preserving and Restoring Neuromuscular
Efficiency

11:30 - 12:30

Maicon Albuquerque

Challenging the Brain During High-Intensity
Exercise

14:30 - 15:30

Camila Lima

Neurophysiological and biomechanical
assessments and neurological treatment in
spastic stroke individuals

16:30 - 17:30

Carlos Cruz-Montecinos

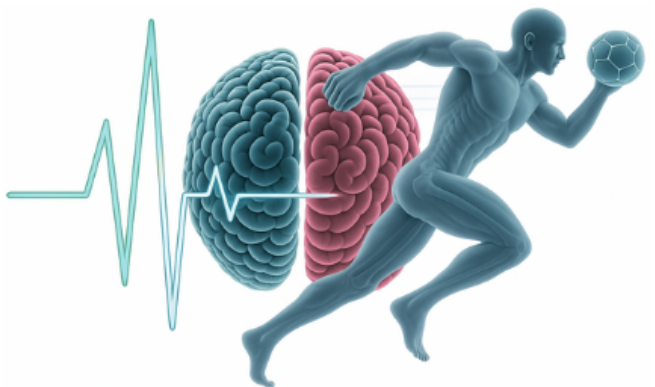
Beyond Chronological Age: Functional
Biomarkers in Aging Models and Their
Association with Multimorbidity and
Geroprotective Agents

17:30 - 18:30

Luis Peñalillo

The Role of Exercise in Muscle Dysfunction in
Chronic Pulmonary Diseases

Day 3



III SNNSEM

III SYMPOSIUM ON NEUROPHYSIOLOGY AND
NEUROMODULATION IN SPORT, EXERCISE
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04/12/25 - 06/12/25



Institute of Basic Health Sciences
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PROGRAM

06/12
(Saturday)

09:30 - 11:00

Anthony Blazeovich

Is the spine greater than the brain? The role of motoneuronal persistent inward currents in motor function and fatigue

11:30 - 12:30

Alexandre Brandão

Gestural interaction with virtual interfaces and kinesiology markers

14:30 - 15:30

Cassio Ruas

The use of transcranial magnetic stimulation (TMS) and transcranial direct current stimulation (tDCS) in exercise science and clinical rehabilitation

16:30 - 17:00

Closing lecture: Guilherme Lage

Neurotechnologies in Motor Behavior

17:30 - 18:30

Closing ceremony



Registration

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04/12/25 -
06/12/25



In-Person
Event



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ABSTRACTS

Category/Topic 1: Exercise Neurophysiology and Neuromodulation



Movement velocity analysis after a rehabilitation program combined with neuromodulation in stroke patients

Marcelo F. A. Manoel¹, Camila D. Lima²

¹Pontifical Catholic University of Campinas, Faculty of Physical Education, University of Campinas

¹(FEF–Unicamp), ²Institute of Physics Gleb Wataghin, UNICAMP

Introduction: Stroke is one of the leading causes of motor disability, frequently resulting in hemiparesis (partial loss of motor function) and spasticity (increased muscle tone and spinal reflexes), affecting their functional performance and leading to the reduction of their movement speed [1]. Movement speed is essential to determine the degree of motor impairment and therefore to monitor functional progress during rehabilitation [3]. The use of cathodal tDCS (c-tDCS) combined with extended reality (XR) exercises, which may decrease spasticity and improve mobility, can lead to restoration and improvement of movement speed [1].

Materials and Methods: Two female participants who suffered ischemic stroke (59 ± 8.48 years, 160 ± 3 cm, 70 ± 2 kg), with the left limb affected, participated in the study through a 10-session rehabilitation protocol for the ankle in combination with c-tDCS. The inhibitory c-tDCS stimulus (1 mA/15 min, Cz) was used during squat exercises performed on top of a ramp with extended reality (XR) (exercises with obstacles above the head). To quantify movement speed, the active ROM was recorded through the Tracker software (São Paulo, Brazil), which assess movement kinematics by identifying ankle joint range of motion without the use of markers pre-to-post c-tDCS. The recorded active ROM data were automatically transferred to Excel. The ROM variation was calculated from the difference between the lowest and the highest value, and later from the highest to the lowest value. After determining the ROM variation, speed was calculated ($\text{Speed} = \Delta\text{ROM} / \Delta t$). The average speed was then computed to determine possible changes. The calculation of the active ROM variation was used to minimize any possible errors in identification from the system.

Results: The active ROM and movement velocity data is presented as mean $\pm$ standard deviation between the two patients. Movement velocity data showed improvements from pre ($3.51 \pm 5.89^\circ/\text{s}$) to-post intervention ($5.36 \pm 14.52^\circ/\text{s}$) with c-tDCS. Active variation ROM was improved for dorsiflexion from the pre-intervention ($3.50^\circ \pm 3.05^\circ$) to post-intervention ($7.04^\circ \pm 4.97^\circ$), while plantarflexion also improved from pre intervention ($-3.67^\circ \pm 7.03^\circ$) to ($-7.45^\circ \pm 4.66^\circ$) to post intervention.

Discussion/Conclusion: The application of c-tDCS associated with the squat exercise on the ramp promoted an increase in ankle movement speed, indicating an improvement in movement execution and efficiency after the intervention. In previous study by Lima et al. (2025) and Manoel et al. (2025) with a similar protocol, the c-tDCS was able to improve spasticity symptoms from stroke participants. In the current study the findings show that c-tDCS contributes to the improvement of active ROM, which can be associated with the improvement in movement velocity since patients had possibly less spasticity after c-Tdcs [1,2,3]. Therefore, the improvement in movement velocity can be associated with better motor control and greater muscular recruitment capacity after 10 sessions of c-tDCS.

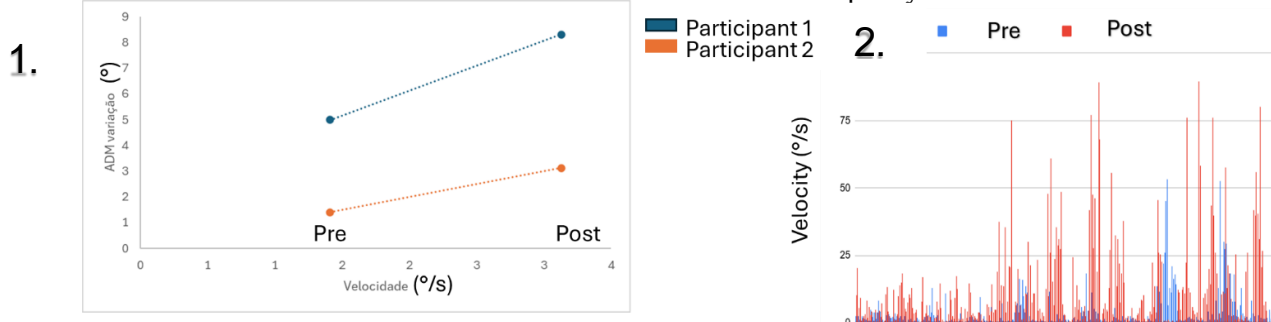


Figure 1. Association between ROM variation and velocity from pre-to post c-tDCS. Figure 2. Velocity from pre-to-post c-tDCS.

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1. LIMA, C.D.; C., RUAS, C.; CARLOS, B.; VASQUEZ, P.; SILVESTRE, J.; BLAZEVOICH, A; CASTELLANO, G. Afferent pathway to the brain in spastic stroke females. ECSS National Meeting, Rimini, ITA, 2025.
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3. RAHMANI, A.; VIALE, F.; DALLEAU, G.; LACOUR, J.-R. Force/velocity and power/velocity relationships in squat exercise. European Journal of Applied Physiology.

EFFECT OF CEREBELLAR TRANSCRANIAL DIRECT CURRENT STIMULATION ON MOTOR LEARNING IN INDIVIDUALS WITH CEREBRAL PALSY

D. L. Ferreira¹, G. M. Lage²

¹ Federal University of Minas Gerais, UFMG.

² School of Physical Education, Physiotherapy and Occupational Therapy, UFMG

Introduction: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique that can enhance motor learning by modulating neuronal excitability. In cerebral palsy (CP), corticothalamic/corticospinal pathways alterations impair motor control and learning. The cerebellum contributes differently to motor adaptation and motor sequence learning, influencing online/offline processes.

Materials and Methods: Randomized, crossover, sham-controlled study. 20 individuals with CP, aged 8–20 years trained on a motor adaptation (MA) or motor sequence (MS) task paired with cerebellar anodal tDCS (1.5 mA, 20 min) or sham. The sequence task required rapid execution of a four-key pattern, while the adaptation task involved unpredictable stimuli. Reaction time (RT), movement time (MT), response time (RSP), errors, and online and offline changes were analyzed.

Results: The tDCS MA group showed improved acquisition, with better performance in Block 10 ($F(1,18)=2.13$; $p=0.02$), faster online RT ($t(18)=-2.06$; $p=0.05$), greater offline MT change ($t(18)=-2.03$; $p=0.05$), and faster overall RT ($F(1,18)=4.06$; $p=0.05$). In contrast, the tDCS MS group demonstrated poorer acquisition, exhibiting slower RT ($F(1,18)=6.04$; $p=0.02$) and reduced RSP ($F(1,18)=5.15$; $p=0.03$).

Discussion/Conclusion: Cerebellar anodal tDCS enhanced learning in motor adaptation but negatively affected motor sequence acquisition. These findings suggest task-dependent effects of cerebellar neuromodulation, benefiting circuits involved in motor adaptation learning while potentially hindering sequential motor processes. This highlights the importance of task-specific stimulation protocols in cerebral palsy rehabilitation.

Effects of Cerebello-Espinal Stimulation Combined with Neurofunctional Rehabilitation in Individuals with Spinocerebellar Ataxia Type 3 Machado Joseph: Protocol Study for a Randomized Clinical Trial

Mariana Brutto de Pinto¹, Liziane Mattos Brião¹, Katine Marchezan Estivalet², Luis Fernando Ferreira^{1,2*}, Fernanda Cechetti¹

¹ Post Graduate Program in Rehabilitation Sciences, Federal University of Health Sciences of Porto Alegre (UFCSPA), Porto Alegre, Brazil.

² Queen's University of Belfast, Belfast, Northern Ireland, United Kingdom.

Introduction: Transcranial direct current stimulation (tDCS) applied to the cerebellum is a promising approach in treatment in spinocerebellar ataxias. Furthermore, rehabilitation is also an effective treatment, however, studies on it are not very detailed. The objective is to describe the protocol of a randomized clinical trial aimed at studying the effects of the combination of tDCS and neurofunctional rehabilitation in the severity of ataxic disease.

Methods: This is a Protocol Study for a Randomized Clinical Trial with two experimental groups, both receiving the Neurofunctional Rehabilitation, consists of trunk strengthening exercises in a seated position, sit-to-stand transitions, Romberg, tandem, and unipedal stances, upper and lower limb coordination activities, standing reach tasks (anterior and lateral), anterior, posterior, and lateral stepping strategies, overground gait with obstacles, direction changes, head movements, and treadmill walking (figure A) - one with real tDCS (figure B) - and the other with sham tDCS totaling 20 sessions for 40 minutes.

Expected Results: The expected outcome is that the study will be safe and feasible.

Discussion: Most studies on physiotherapy do not specify the exercises; therefore, a detailed study of the protocol is necessary for external validation.

	*Obs: the models in the photos are the researchers 
Figure A- Neurofunctional Rehabilitation Protocol	Figure B- TDCS cerebello-spinal

Relation between joint power and motor unit discharge properties

Leonardo M. Silva¹, Ricardo P. Schiochet¹, Anderson S. Oliveira², Francesco Negro³, Guilherme R. Silva¹, Fabrizio Caputo¹

¹Human Performance Research Group, UDESC, ²Department of Materials and Production, AAU, ³Department of Clinical and Experimental Sciences, UNIBS

Introduction: Power is essential in a wide range of movements and in most sports. It is known that neural factors play a crucial role in muscle power production, as higher movement velocities generate greater electromyographic activity amplitude, depending on the context. Assessing motor-unit behavior during muscle shortening and lengthening using high-density surface electromyography (HD-sEMG) can provide a deeper understanding of the mechanisms underlying power production. Therefore, this study aims to determine the association between maximal power (Pmax) and the firing properties of motor units (MU) assessed using HD-sEMG during dynamic and isometric contractions.

Materials and Methods: In session 1, 20 men performed maximal isotonic knee extensions at 4–5 loads that allowed lift average velocities of 0.3-1 m/s, followed by a one-repetition maximum (1-RM) determination. Pmax was derived from the mean angular velocity (AV) and mean power using a second-order polynomial regression, accounting for the theoretical zero and theoretical maximal AV, and then relativized by body mass. In session 2, the same participants performed a maximal voluntary isometric contraction (MVC), followed by separate tasks of isometric and slow movement cycles (5 %/s) of knee extensions at 30% and 70% of their respective maximal capacities (MVC or 1-RM). HD-sEMG was acquired from the left vastus lateralis with a 64-electrode grid. The root mean square (RMS) was calculated throughout the concentric phase, then the RMS slope (sRMS) was determined. In addition, all EMG data were decomposed using convolutive blind source separation. For both exercise intensities (30% and 70%), the slopes of discharge rate during the concentric phase (sDR) were calculated, as were the peak discharge rates (iDR-P) for the isometric tasks. Finally, Pearson correlations were used to evaluate the associations between Pmax and sRMS, sDR, and iDR-P.

Results: During dynamic contractions, Pmax showed a significant positive correlation with sDR at 70% 1-RM ($r = 0.77$, $p < 0.05$), but not at 30% 1-RM ($r = 0.47$, $p > 0.05$). There was no significant correlation between Pmax and sRMS at 30% and 70% 1-RM ($r = 0.003$, $p > 0.05$; and $r = 0.36$, $p > 0.05$, respectively). Furthermore, iDR-P showed no significant correlation with Pmax at 30% ($r = 0.25$, $p > 0.05$) and 70% ($r = -0.007$, $p > 0.05$) MVC in isometric contractions.

Discussion/Conclusion: The sDR represents the rate of change in neural drive, and its significant association with Pmax at 70% 1-RM suggests that ballistic performance may be related to motor units' ability to increase neural drive during high-intensity muscle shortening. Furthermore, the lack of association between Pmax and global EMG variables and isometric firing rate properties highlights the limitations of these parameters in explaining human performance.

Motor unit behavior from vastus lateralis when comparing isometric and dynamic contractions at similar relative contraction intensities

Ricardo P. Schiochet¹, Guilherme R. Silva¹, Anderson S. Oliveira², Francesco Negro³, Leonardo M. Silva¹, Fabrizio Caputo¹

¹Human Performance Research Group, UDESC, ²Department of Materials and Production, AAU, ³Department of Clinical and Experimental Sciences, UNIBS

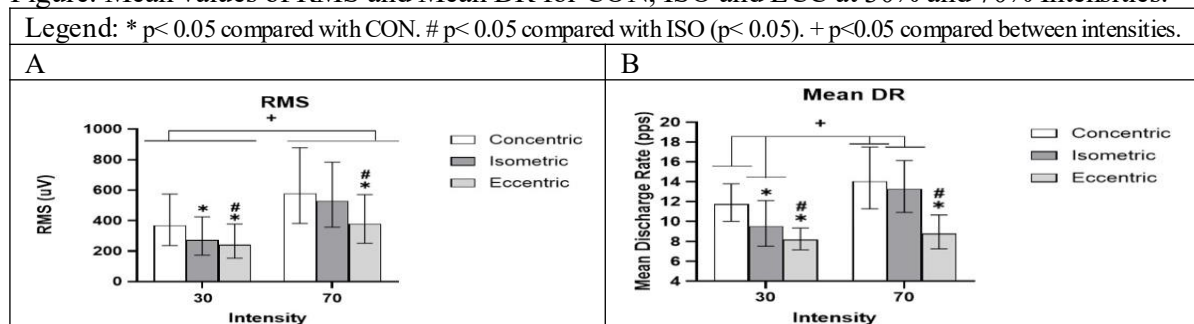
Introduction: Muscle force production results from neuromechanical factors that differ across contraction types, creating a maximum torque hierarchy: eccentric (ECC) > isometric (ISO) > concentric (CON). Advances in high-density surface electromyography (HD-sEMG) decomposition allow for an in-depth investigation of motor unit (MU) neural modulation strategies. Therefore, the aim of this study is to compare the behavior of the same MUs between similar relative contractions, at 30% and 70% of maximal isometric and dynamic torque, to determine the effect of contraction type on the neural drive to muscles.

Materials and Methods: Eighteen active men (n=18) performed knee extensions during isometric (ISO) (60°) and dynamic (50° to 70° at 5°/s) contractions, normalized to maximal isometric and dynamic torque (MIT/MDT), at 30% and 70%. The HD-sEMG signals (vastus lateralis muscle, 64-channel grid) were decomposed and the identified motor units were tracked across contraction modes. Linear mixed models tested the effect of contraction type and intensity on the root mean square (RMS) and mean discharge rate (M-DR).

Results: CON showed a significant 1.33 and 1.21-fold increase in RMS and M-DR ($p < 0.001$; $d = 0.64$; $p < 0.001$; $d = 1.61$), respectively, relative to ISO at 30%, whereas at 70% the CON values of 1.09-fold for RMS and 1.04-fold for M-DR showed no significance in the comparison ($p = 0.170$; $p = 1.000$).

Discussion/Conclusion: Finally, the results suggest that at 30%, shortening contractions exhibit distinct myoelectric properties, possibly due to mechanical differences of the exercise. At 70%, however, a similar behavior was observed, consistent with a possible saturation of MUs, indicating that mechanical components may influence the neural drive more at low intensities than at high ones. Global Muscle activity, measured as RMS, matched well with the discharge rate behavior of the decomposed MUs.

Figure: Mean values of RMS and Mean DR for CON, ISO and ECC at 30% and 70% Intensities.



Effects of a Deep-Water Walking Training on Cognition in Older Adults: A Quasi-Experimental Study

W. T. Borba¹, A. T. G. Moreira², J. C. Zanella², K. P. Bitencourt², A. C. Koenig², A. C. Kanitz²

¹Grupo Multidisciplinar de Pesquisa em Exercício Físico e Saúde, GEFiS, ²Centro de Referência do Envelhecimento e Movimento, CREM, ³Universidade Federal do Rio Grande do Sul, UFRGS

Introduction: Aging is associated with cognitive declines that affect attention, processing speed, and executive functions. Physical exercise is an effective non-pharmacological strategy to preserve cognition, and deep-water walking, by combining motor and sensory stimuli, presents potential cognitive benefits. To the best of our knowledge, no previous studies have evaluated the effects of this modality on cognitive aspects. Therefore, this study aimed to analyze the effects of a deep-water walking training program on cognition in older adults.

Materials and Methods: This quasi-experimental study used data from a community-based deep-water walking project conducted at the Reference Center for Aging and Movement (CREM/UFRGS). A total of 71 older adults (≥ 60 years) participated in 2024, completing four months of training, twice per week, with 45-minute sessions. Training intensity, composed of continuous and interval aerobic exercise, was monitored using the Borg Rating of Perceived Exertion Scale (6–20). Cognition was assessed using the Montreal Cognitive Assessment (MoCA), analyzing total scores and seven cognitive domains (visuospatial, naming, memory, attention, language, abstraction, delayed recall, and orientation). Descriptive (means, standard deviations, frequencies) and inferential analyses (paired t-test) were conducted using JASP software (v.0.95.2.0), with a significance level of $\alpha = 0.05$.

Results: Of the 71 older adults enrolled, 60 completed the program, resulting in 84.5% adherence, with an average attendance rate of 68.5%. The total MoCA score remained stable between the pre-test (23.42 ± 3.87) and post-test (23.60 ± 4.11) with no significant difference ($p = 0.763$). At baseline, 8 participants (11.3%) showed cognitive decline, and 9 (12.7%) at the end. Analysis of cognitive domains also showed no significant differences between pre- and post-intervention: visuospatial ($p = 0.829$), naming ($p = 0.687$), attention ($p = 0.197$), language ($p = 0.416$), verbal fluency ($p = 0.133$), abstraction ($p = 0.094$), delayed recall ($p = 0.745$), and orientation ($p = 0.293$).

Discussion/Conclusion: These results indicate maintenance of global cognitive function and of all MoCA domains after the intervention period. The stability of the findings suggests that deep-water walking may serve as an effective strategy to preserve cognition in older adults. Long-term follow-up studies are recommended to further explore the cognitive effects of this aquatic exercise modality.

Effects of a community-based aquatic exercise program on cognition in older adults: a quasi-experimental study

A. J. T. Jacques¹, W. T. Borba², A. T. G. Moreira², J. C. Zanella², K. P. Bitencourt², A. C. Kanitz²

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Introduction: Aging is a natural process that leads to physiological changes, such as cognitive decline. In turn, regular physical exercise is an excellent tool to combat the adverse effects of this process, as it induces neuroplasticity important for maintaining autonomy and cognitive function. In this context, water-based exercise appears to be an interesting modality for cognitive health, as it involves tasks that require participants' motor coordination and attention. Thus, the purpose of this study was to evaluate the effects of a water-based exercise program on the cognitive aspects of community-dwelling older adults.

Materials and Methods: This was a quasi-experimental study with independent older adults (≥ 60 years) participating in a community aquatic exercise program. The training consisted of water-based exercise, prescribed in an aerobic and interval format. The frequency was two sessions per week, each lasting 45 minutes, totaling four months of intervention. The test used to evaluate cognition was the Montreal Cognitive Assessment (MoCA) with its respective domains: visuospatial/executive, naming, attention, calculation, language, verbal fluency, abstraction, delayed recall, and orientation. The test has a score of 0 to 30 points, and a cutoff point of 19 was used to define the presence of cognitive decline. The test was applied before and after the program. Inferential tests (Wilcoxon Test) were performed using the JASP software, adopting a significance level of $\alpha = 0.05$.

Results: A total of 52 older adults (two men and 50 women) were included in the project, and 42 completed the intervention, resulting in an average adherence of 82 ± 18 %. The number of individuals classified with cognitive decline was three at pre-intervention and five at post-intervention. The comparison of the total mean MoCA score did not show significant differences ($p = 0,666$) in cognitive performance (pre: $23,42 \pm 3,87$; post: $23,60 \pm 4,11$). The domain analysis also showed no significant changes: visuospatial/executive ($p = 0,478$), naming ($p = 0,464$), attention ($p = 1,000$), calculation ($p = 0,802$), language ($p = 0,881$), verbal fluency ($p = 1,000$), abstraction ($p = 0,741$), delayed recall ($p = 0,278$), and orientation ($p = 0,246$).

Discussion/Conclusion: The participants presented a cognitive profile without impairment already at the pre-intervention moment. The intervention demonstrated the ability to maintain cognitive levels, even with a low-frequency community aquatic exercise program. Therefore, water-based exercise presents as a promising tool in the prevention of cognitive decline associated with aging.

BENEFITS OF STRENGTH TRAINING FOR COGNITIVE FUNCTIONS

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Introduction: The practice of physical exercise, especially strength training, offers well-recognized benefits in the areas of health, aesthetics, and physical performance. Despite the evidence on physiological and bodily effects, there is still a scarcity of studies addressing the impact of strength training on cognitive processes. This study aimed to investigate the effects of strength training on cognition, with a particular focus on executive functions.

Materials and Methods: For this purpose, a software with tasks to measure reaction time was used to assess cognitive performance in two groups: one composed of individuals who had been practicing strength training for at least two years, and another composed of individuals who had never practiced it. Measures of reaction time (RT) and number of errors (NE) in inhibition tasks were analyzed.

Results: The results showed that, for RT, the t-test did not indicate significant differences between groups. However, the Mann–Whitney U test revealed a significant difference in NE in inhibition tasks ($p \leq 0.05$), indicating that the training group had fewer errors compared to non-practitioners.

Discussion/Conclusion: It is concluded that strength training may exert a beneficial effect on certain aspects of cognitive function. The results reinforce the importance of investigating which cognitive processes are most sensitive to regular strength training, contributing to a better understanding of the relationship between physical exercise and cognition.

ABSTRACTS

Category/Topic 2: Neuromuscular Adaptations to Exercise



MUSCLE SWELLING OF THE VASTUS LATERALIS DURING A SPRINT INTERVAL TRAINING SESSION

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Introduction: Sprint Interval Training (SIT) is a conditioning method used in a variety of scenarios, from disease prevention and treatment to sport training. It is recognized that an acute increase in muscle thickness (MT) during physical effort may indicate metabolic stress and serve as an indicator of fatigue. Therefore, the aim of this study was to quantify quadriceps MT throughout a SIT session. **Materials and Methods:** Three participants (2 men: age 22.5±0.71 years; weight 82 ±14.14 kg; height 179±14.14 cm, and 1 woman: age 21 years; weight 73 kg; height 167 cm) performed 6 maximal sprints on a cycle ergometer (Ergofit 167 Cycle, Germany), consisting of 30s all-out efforts followed by 4 min of passive recovery. Participants maintained a cadence ≥120 rpm with a self-selected load until voluntary exhaustion. MT measurements were performed using A-mode ultrasound (Bodymetrix; Intelametrix, USA) on the vastus lateralis, in which the ultrasound trajectory ranged from 66% to 90% of the line connecting the anterior superior iliac spine to the lateral edge of the patella. Measurements were taken with participants seated and knees at a 90° angle. For inferential analysis, Generalized Estimating Equations were applied, with Bonferroni post-hoc tests for moment comparisons and a significance level of $p<0.05$. **Results:** The analysis showed differences in MT between moments ($p<0.01$); however, the difference occurred only in the initial measurements (table 1). There was no significant difference between pre- and post-sprint evaluations ($p=1$), although differences were found between M0 and M1, as well as between M0 and M2 ($p<0.01$ and $p=0.034$, respectively). **Discussion/Conclusion:** A systematic review by Volland et al. (2017) suggests that a low number of sprints in an SIT session appears to be sufficient to improve physical fitness. The initial reduction in MT may be explained by the redistribution of blood flow during the first sprints, followed by swelling from the fourth effort onward. The SIT protocol was not sufficient to cause an acute increase in MT, and the self-selected load may have underestimated effort during the tests.

Table 1. Muscle thickness (mm) x sprint number.

Moments	M0	M1	M2	M3	M4	M5	M6
Male 1	24,32	18,87	20,43	22,11	-	-	-
Female 1	29,53	27,48	26,20	30,48	28,02	31,26	30,53
Male 2	26,65	23,62	26,04	25,22	30,70	28,23	28,34
Mean	26,83a	23,32b	24,22bc	25,94ac	29,36a	29,75a	29,44a
SD	±2,61	±4,31	±3,29	±4,23	±1,89	±2,14	±1,54

Different letters indicate statistically significant differences between time points. M0 = pre-intervention, M1 = after the first sprint, etc.

The shear modulus behavior of biceps femoris long head and medial gastrocnemius are intensity-specific during isometric knee flexion

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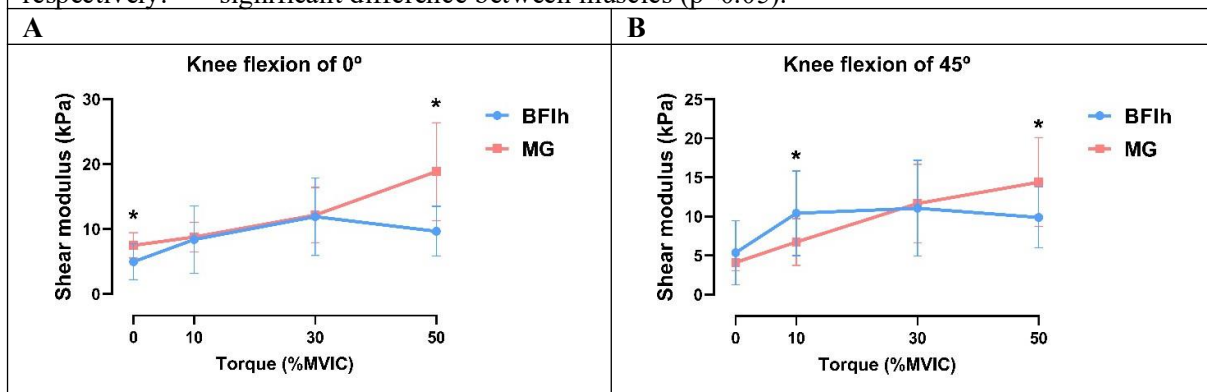
Introduction: We explored the shear modulus behavior of biceps femoris long head (BF_{lh}) and medial gastrocnemius (MG) in different knee flexion angles at rest and during submaximal isometric contractions.

Materials and Methods: Twenty physically active males (21.9 ± 2.4 years, 77.7 ± 11.6 kg, 170 ± 1.0 cm) provided informed consent to participate in this ethics-approved study. Participants attended three sessions. On day 1, questionnaires were completed, familiarization was conducted, and maximal voluntary isometric contraction (MVIC) of knee flexion was assessed using an isokinetic dynamometer with the knee and hip at 30° and 0° , respectively. Submaximal target torques of 10%, 30%, and 50% of MVIC were derived. On days 2 and 3, the shear modulus of the BF_{lh} and MG was measured at rest and at 10%, 30%, and 50% of MVIC, with the knee flexed (45°) and extended (0°). Data normality was verified using the Shapiro–Wilk test, and differences in shear modulus between BF_{lh} and MG were analyzed using paired t-tests or Wilcoxon tests ($\alpha = 0.05$).

Results: With the knee extended, the MG exhibited a higher shear modulus than the BF_{lh}, by 51% at rest ($p < 0.001$) and 95% at 50% of MVIC ($p < 0.001$; Figure 1A). With the knee flexed at 45° , BF_{lh} showed a 54% greater shear modulus than MG at 10% of MVIC ($p < 0.001$), whereas MG demonstrated a 45% higher value at 50% of MVIC ($p < 0.001$; Figure 1B).

Discussion/Conclusion: Assuming the shear modulus reflects the muscle's elastic modulus, and therefore its tensile stiffness during contraction, these findings suggest that the relative contribution of each muscle to torque production depends on contraction intensity and knee flexion angle.

Figure 1: Biceps femoris long head (BF_{lh}) and medial gastrocnemius (MG) shear modulus at rest, 10, 30 and 50% of the maximum voluntary isometric contraction (MVIC) with the knee extended (A) and flexed at 45° (B). Red squares and blue filled circles represent the MG and BF_{lh} muscles, respectively. * = significant difference between muscles ($p < 0.05$).



Effects of a 20-week physical exercise program on mobility, balance, and fear of falling in older adults

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Aging is associated with decreased mobility and balance, factors that contribute to an increased risk of falls and fear of falling. This study aimed to evaluate the effects of a 20-week physical exercise program on these variables in older adult participants from the e Reference Center for Aging and Movement (CREM) in 2022. A total of 166 older adults participated (mean age: 73.28 ± 6.88 years), including 144 women (83%) and 29 men (17%). The Timed Up and Go (TUG) test for functional mobility, the one-leg stance test with eyes open and closed, and the FES-I scale to assess fear of falling were applied at pre- and post-intervention moments. Significant improvements were found in mobility (pre: 6.7 ± 1.7 s; post: 6.3 ± 1.7 s; $p=0.001$) and a significant reduction in fear of falling (pre: 22.92 ± 8.70 ; post: 20.35 ± 9.01 ; $p=0.001$). One-leg stance balance, both with eyes open ($p=0.203$) and with eyes closed ($p=0.179$), did not show significant changes. In conclusion, the physical exercise program improved functional mobility and reduced fear of falling, indicating important benefits for the autonomy and confidence of older adults in daily activities.

Muscle size is not explains everything: quality also matter on performance

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Introduction: Morphological aspects of skeletal muscle tissue have important influence regarding strength performance. The present objective of this study is to understand how much of performance from futsal athletes is explained by the skeletal muscles of lower limbs. **Materials and Methods:** Twenty-nine futsal athletes (22.03 ± 3.25 years; 77.14 ± 9.42 kg; 175.13 ± 7.05 cm) were evaluated (number: 5.698.140). Firstly, athletes were positioned in prone above a stretcher for 10 minutes aiming body fluid redistribution. After, an experienced sonographer performed two quadriceps and hamstrings cross-sectional area images at 30%, 40% and 50% of femur length using the extend field of view mode from an ultrasound. Moreover, there was performed two longitudinal extended field-of-view images of the patellar tendon. Subsequently, athletes were positioned in an isokinetic dynamometer chair with 85° of hip flexion aligning knee and dynamometer rotation axis. Warm-up started with 20 concentric knee extension/flexion contractions at 120°/s. After, five concentric knee extensions at 60 and 300°/s, finishing the protocol with five eccentric knee flexions contractions at 60°/s, only dominant lower limb was evaluated, with two minutes of rest between series. Athletes also performed three countermovement jumps (CMJ) using two force plates. For kinetic analyses, the torque peak from concentric and eccentric contraction was registered, and CMJ height was obtained using and custom-made Python script using ground reaction force. Ultrasound images were analyzed using ImageJ® software, obtaining cross-sectional area (CSA) from quadriceps and hamstring muscles separately, muscle quality was also assessed using histogram mode and corrected by the adipose tissue above the interested muscle. Tendon thickness, quality and length were analyzed using the same software. A backward linear multiple regression model was used to determine the explanation of dependent variables (concentric knee torque at 60 and 300°/s, eccentric knee flexion torque and CMJ height) by independent variables (muscle CSA and quality; tendon thickness, length, and quality). **Results:** Vastus lateralis muscle quality and CSA, vastus intermedius quality, patellar tendon thickness, length, and quality explain 41.5% and 31.9% of concentric torque at 60 and 300°/s, respectively (OB=0.66–0.85; ES=0.46–0.70; $p < 0.05$). Semimembranosus CSA and quality explain 33.1% of eccentric torque (OB=0.69; ES=0.49; $p < 0.05$). Vastus lateralis muscle quality and CSA, vastus intermedius quality, patellar tendon thickness, length, and quality explain 45.6% of CMJ height (OB=0.91; ES=0.83; $p < 0.05$). **Discussion/Conclusion:** The present results may be explained since muscle quality indicates the amount of contractile tissue present in the muscle belly, which has great relevance regarding muscle strength. Tendon characteristics as it thickness and quality (e.g. more collagenous fibers and better aligned), also contribute to the strength transmission produced by the muscle. In sum, morphological and quality from different tissues are determinants for performance.

Diferenças nos Componentes da Aptidão Funcional entre Níveis de Mobilidade em Idosos Longevos

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Introdução: O presente estudo teve como objetivo analisar as diferenças nos componentes da aptidão funcional (força, flexibilidade, equilíbrio e resistência cardiorrespiratória) entre níveis distintos de mobilidade em pessoas idosas longevas (80 anos ou mais) participantes de um projeto de extensão universitária. Um objetivo adicional foi identificar quais componentes da aptidão funcional melhor discriminam os diferentes níveis de desempenho na mobilidade, contribuindo para a compreensão dos fatores determinantes da funcionalidade e autonomia em idades avançadas.

Material e Métodos: Estudo transversal com 66 idosos longevos (≥ 80 anos; $84,5 \pm 3,4$ anos), todos em condição de destreino, participantes de um projeto de extensão universitária. Avaliaram-se a mobilidade funcional (Timed Up and Go – TUG) e os componentes da aptidão funcional por meio do *Senior Fitness Test*: força de membros inferiores e superiores, flexibilidade, equilíbrio e resistência cardiorrespiratória. Os participantes foram divididos em quartis conforme o tempo no TUG (Q1 = mais rápido; Q4 = mais lento). As comparações entre grupos foram realizadas pelo teste de Kruskal–Wallis com pós-teste de Dunn (correção de Holm), e os contrastes extremos (Q1×Q4) pelo teste de Mann–Whitney. O tamanho de efeito foi calculado por ε^2 e r .

Resultados: Houve diferenças significativas entre os quartis na força de membros inferiores, equilíbrio e resistência cardiorrespiratória. A força de membros inferiores apresentou melhor desempenho no Q1 em relação a Q3 e Q4 ($H(3)=11,79$; $p=0,008$; $\varepsilon^2=0,197$), com efeito forte no contraste Q1×Q4 ($r=0,61$). O equilíbrio foi o componente mais discriminativo ($H(3)=35,55$; $p<0,001$; $\varepsilon^2=0,573$), enquanto a resistência cardiorrespiratória também diferiu entre quartis ($H(3)=16,26$; $p=0,001$; $\varepsilon^2=0,280$), favorecendo os idosos mais ágeis. Não foram observadas diferenças significativas na flexibilidade e na força de membros superiores. Nos contrastes extremos, a força de membros inferiores ($d=-1,39$) e o equilíbrio ($d=-2,68$) apresentaram os maiores tamanhos de efeito, indicando que esses componentes melhores distinguem os níveis de mobilidade funcional.

Discussão e Conclusão: Mesmo em condição de destreino, os idosos com 80 anos ou mais apresentaram diferenças marcantes na força de membros inferiores e no equilíbrio entre os diferentes níveis de desempenho funcional. Esses achados indicam que tais componentes são determinantes para a mobilidade e independência nessa faixa etária. Além disso, o bom desempenho funcional parece estar mais associado à manutenção da aptidão física do que à idade cronológica em si, reforçando a importância de programas de exercício voltados ao fortalecimento muscular e ao treino de equilíbrio para promover a autonomia e a qualidade de vida dos longevos.

Association between cognition and functional fitness in the elderly at CREM

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Introduction: The aging process is accompanied by progressive changes in physical and cognitive functions that can compromise independence and quality of life in older adults. Evidence indicates that cognitive performance and functional fitness are interrelated, as both depend on neural, muscular, and cardiovascular integrity. Understanding how cognitive function relates to the components of functional fitness can contribute to identifying key factors for maintaining autonomy and healthy aging. Therefore, the objective of this study was to analyze the relationship between cognitive performance and components of functional fitness in older adults participating in an extension program. **Materials and Methods:** This cross-sectional study analyzed baseline data from participants enrolled in 2023 (n=123; 96 women and 27 men) and 2024 (n=6; 3 women and 3 men) at the *Centro de Referência em Envelhecimento e Movimento* (CREM), totaling 129 older adults (mean age: 69.45 ± 6.03 years; 99 women and 30 men). Inclusion criteria were: age ≥ 60 years, participation in the CREM evaluation protocol, and absence of diagnosed neurodegenerative diseases or cognitive impairments that could interfere with testing. All participants underwent standardized assessments of cognitive performance and functional fitness, following the center's established procedures. Physical fitness was assessed using the *Senior Fitness Test* battery, which included the 30-second Chair Stand Test to evaluate lower-limb strength (FMI), the 30-second Arm Curl Test to assess upper-limb strength (FMS), and the 2.44-meter Up-and-Go Test (EQ2.44) to measure balance and agility. Cognitive performance was assessed using the Portuguese version of the *Montreal Cognitive Assessment* (MoCA). Associations between functional performance and cognition were analyzed using Spearman's correlation coefficient, with a significance level set at $p \leq 0.05$. Statistical analyses were performed using JASP software (version 0.19.3). **Results:** A moderate positive correlation was observed between MoCA scores and lower-limb strength (FMI) ($\rho = 0.316$; $p \leq 0.001$), and a moderate negative correlation between MoCA and balance/agility (EQ2.44) ($\rho = -0.354$; $p \leq 0.001$). A weak positive correlation was also found between MoCA and upper-limb strength (FMS) ($\rho = 0.210$; $p \leq 0.05$). **Discussion and Conclusion:** Cognitive performance was found to be associated with lower- and upper-limb strength, as well as balance and agility in untrained older adults. The observed correlations indicate that cognition exerts a global influence on physical performance, although with varying magnitudes across different components of functional fitness. These findings highlight the importance of integrating cognitive and physical dimensions in strategies aimed at promoting healthy aging.

Effects of a 12-week dance intervention on cognition and physical aspects in older adults: a community-based experimental study

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Introduction: Population aging challenges the maintenance of brain health and functional capacity. Dance, by integrating physical, social, and cognitive components, is a promising non-pharmacological intervention to promote active and healthy aging. The aim of this study was to evaluate the effects of a 12-week dance program on global cognition (GC) and physical aspects (lower limb strength – LLS, and cardiorespiratory endurance – CE) in older adults.

Materials and Methods: A pilot study was conducted with a 12-week dance program. The sample consisted of 19 older adults (age ≥ 60 years) who participated in dance classes twice a week and were assessed before and after the intervention. Global cognition was evaluated using the Montreal Cognitive Assessment (MoCA), cardiorespiratory endurance (CE) with the 6-Minute Walk Test (6MWT), and lower limb strength (LLS) with the 30-Second Chair Stand Test and the Five Times Sit-to-Stand Test. Pre- and post-intervention data were compared using either the paired t-test or the Wilcoxon test, and associations between variables were analyzed using Pearson's correlation, with a significance level of 5%.

Results: There was no statistically significant change in global cognition ($p = 0.500$) or lower limb strength ($p > 0.05$). However, cardiorespiratory endurance improved significantly, with a mean increase of 38.3 meters in the 6-Minute Walk Test ($p = 0.003$). Post-intervention correlation analysis revealed a positive and moderate association between cardiorespiratory endurance and global cognition ($r = 0.518$, $p = 0.023$).

Discussion/Conclusion: A 12-week dance program improved cardiorespiratory endurance in older adults. Although it did not significantly change global cognition, the positive correlation found between cardiorespiratory endurance and global cognition suggests that improvements in cardiorespiratory fitness may be associated with better cognitive performance, indicating a relevant avenue for future research.

Strength Training and Functional Performance in Older Adults: A Pre- and Post-Intervention Assessment of IADL Performance

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This study aims to analyze the associations between muscle strength and instrumental activities of daily living (IADL) among older adults participating in the Municipal Reference Center (CREM) in 2022, before and after 20 weeks of physical exercise. Handgrip strength (FPM), lower limb strength (FMI), the five times sit-to-stand test (FMI5X), and the total IADL scale score were evaluated to verify how muscle strengthening impacts functional autonomy. The results showed that at baseline (pre-intervention), there was a significant correlation between IADL and FMI ($r = 0.324$; $p < 0.001$), indicating that greater lower limb strength is associated with greater autonomy in daily activities, and a negative correlation between IADL and FMI5X ($r = -0.381$; $p < 0.001$), suggesting that a shorter time on the test is associated with higher functional performance. In terms of means, the pre-intervention FPM was $19.616 (\pm 6.928)$ and increased to $23.458 (\pm 5.816)$ after 20 weeks of exercise. The FMI had a mean of $14.796 (\pm 4.205)$ at the beginning of the study, increasing to $16.108 (\pm 4.925)$ at the end. The time on the FMI5X test was $10.171 (\pm 3.262)$ at pre-test and decreased to $9.604 (\pm 3.278)$ post-training. The total IADL score had a mean of $26.547 (\pm 1.189)$ before the intervention and $26.379 (\pm 1.843)$ after the 20 weeks, with no statistically significant difference between the periods ($t = 1.268$, $p = 0.207$). Post-intervention, a correlation was observed between IADL and FMI5X ($r = -0.296$; $p < 0.001$), suggesting a reduced direct dependence of strength on functional autonomy. The paired t-test indicated significant increases in the FPM ($t = -13.649$, $p < 0.001$) and FMI ($t = -4.745$, $p < 0.001$) variables, and a significant decrease in the FMI5X test time ($t = 4.656$, $p < 0.001$), whereas the IADL score showed no significant alteration ($t = 1.268$, $p = 0.207$). These results suggest that, although the increase in muscle strength is relevant, the direct impact of strength on autonomy decreases post-program, indicating a more efficient balance between muscular capacity and functional performance, with the older adults managing to improve their autonomy even without a direct dependence on strength.

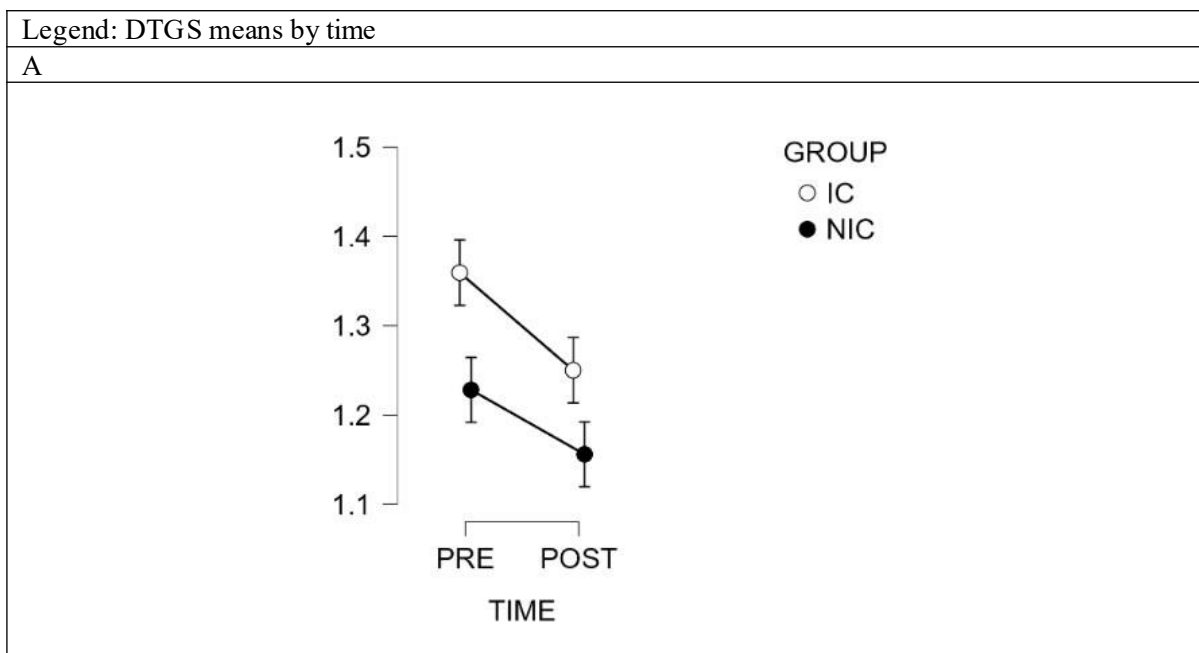
Effects of physical exercise in dual-task walking speed in older adults

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Introduction: The aging process leads to changes in organism that affects physical and cognitive capacities. Gait is a important aspect of functionality because walking means independence. The ability to perform motor-cognitive tasks may serve as an indicator of changes in global cognitive function. Physical exercise emerges as a potential intervention to enhance these capabilities. The objective of this study was to evaluate the effect of physical exercise on dual-task gait speed in older adults with and without cognitive impairment. **Materials and Methods:** A total of 192 older adults (aged 60 years or older) were included in this study. These individuals were participants in an extention program affiliated with ESEFID/UFRGS, specifically the Center for Reference in Aging and Movement (CREM). All participants engaged in regular physical exercise twice a week for 50 minutes per session. The activities included Deep Water Walking, Dance, Balance Training, Aquatic Physiotherapy, Gymnastics, Hidrohymnastics, and Resistance Training. Based on their Montreal Cognitive Assessment (MoCA) scores, participants were categorized into two groups: Non-Impaired Cognitively Group (NIC; n=123) and Cognitively Impaired Group (IC; n=68). Self-selected gait speed under dual-task conditions (DTGS) was assessed using the 10-Meter Walk Test (10MWT), incorporating a continuous subtraction cognitive task. Statical analysis were conducted by an ANOVA two-way of repeated measures with Bonferroni post-hoc ($p \leq 0,05$). **Results:** The mean ages of the NIC and IC groups were, respectively $74,49 \pm 7,38$ e $71,59 \pm 5,97$. The results revealed a statistically significant difference in the factor Time ($p \leq 0,001$) and Group ($p = 0,008$). The means of DTGS in NIC group were $1,22 \pm 0,27$ m/s in PRE and $1,15 \pm 0,35$ m/se in POST ($p = 0,017$). In IC group were $1,35 \pm 0,29$ in PRE and $1,25 \pm 0,28$ in POST ($p = 0,005$). **Discussion/Conclusion:** Physical exercise represents a critical intervention for preserving individual autonomy, owing to its impact on both physical and cognitive capacities. This study demonstrated that performance in dual-tasking abilities can be enhanced through structured and supervised exercise protocols.

Figure/Table/Graph:



Relação entre força de membros inferiores e equilíbrio unipodal em idosos com diferentes níveis de aderência ao exercício

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INTRODUÇÃO: A relação entre força de membros inferiores e equilíbrio unipodal é fundamental para a funcionalidade e prevenção de quedas em pessoas idosas. Aspectos como condição física, percepção corporal e medo de cair influenciam o desempenho. Considerando que a modalidade de equilíbrio visa desenvolver força, estabilidade postural e autonomia, compreender a interação entre estas capacidades é importante para aprimorar programas de exercícios no envelhecimento.

OBJETIVO: Comparar o desempenho de força muscular de membros inferiores e de equilíbrio entre pessoas idosas com diferentes níveis de aderência a um programa de exercícios físicos voltado ao equilíbrio, e verificar a associação entre a força muscular e o equilíbrio corporal.

MÉTODOS: Trata-se de um estudo quase-experimental com 14 idosos ($77 \pm 7,5$ anos) participantes da modalidade de equilíbrio de um programa de extensão universitária (CREM/UFRGS), divididos em 2 grupos conforme a aderência (G1: maior e G2: menor). Foram aplicados testes de força Sentar e Levantar por 30 segundos (TSL30') e em 5 repetições (TSL5x) e os de equilíbrio, teste de apoio unipodal com olhos abertos (EUOA) e olhos fechados (EUOF), aplicados na pré e pós-intervenção de seis meses. A análise estatística descritiva, teste t para amostras independentes, ANOVA mista de medidas repetidas e correlação de Pearson ($p < 0,05$) no software JASP.

RESULTADOS: Em todas comparações entre grupos e intervenção não houve diferença estatisticamente significativa ($p > 0,05$). Houve somente efeito do tempo no TSL5x ($F(1,12)=9.88$; $p=0.008$; $\eta^2=0.45$) e no EUOA ($F(1,12)=5.60$; $p=0.036$; $\eta^2=0.32$). Também não foi identificada associação significativa entre os testes de força e de equilíbrio (todos $p > 0,05$). As correlações intra-grupos: no G1, $r=0,38$ ($p=0,35$) entre $\Delta TSL5x$ e $\Delta EUOA$, e no G2, tendência positiva entre $\Delta TSL30'$ e $\Delta EUOF$, $r=0,70$ ($p=0,12$), porém sem significância estatística.

CONCLUSÃO: Os resultados sugerem que força e equilíbrio podem apresentar adaptações independentes em pessoas idosas, mesmo diante de um programa estruturado de exercícios e níveis distintos de aderência. Estudos com amostras maiores e maior controle da intensidade e frequência dos treinos são recomendados para esclarecer melhor a interação entre força e equilíbrio em participantes da modalidade de equilíbrio.

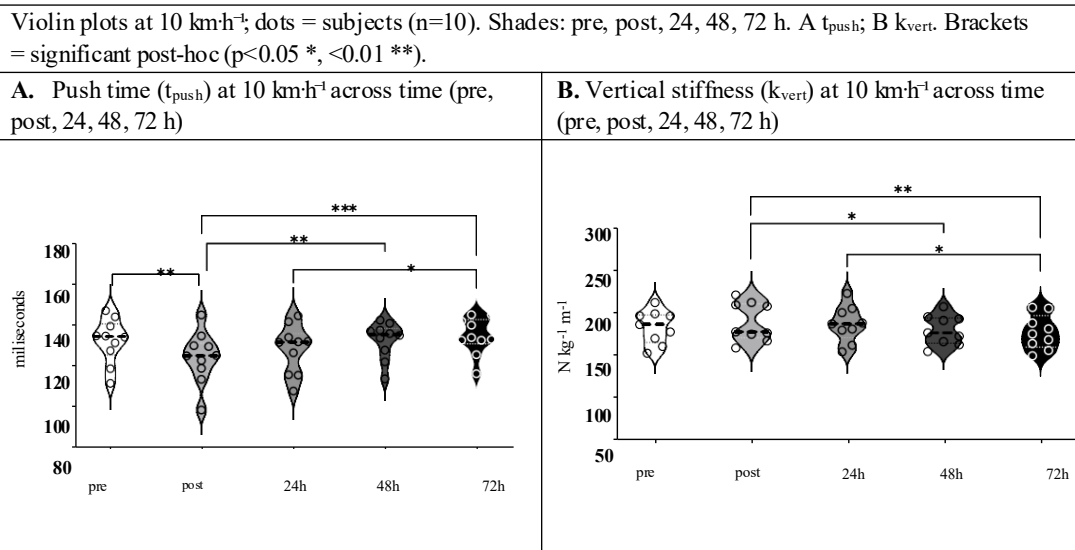
Palavras-chave: Idoso. Equilíbrio. Força.

Spring–Mass Rebound in Running after Muscle Damage

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Introduction: The spring–mass rebound depends on tendon–muscle contributions, reflected in landing-to-take-off timing. t_{brake} is the time while center of mass energy decreases (negative work) and t_{push} the time while it increases (positive work). A longer t_{push} than t_{brake} signals deviation from a spring–mass rebound. Vertical stiffness (k_{vert}) rises when tendon contribution increase. Exercise-induced muscle damage (EIMD) may shift these timings. **Material and Methods:** Ten healthy men (29.4 ± 2.7 y) performed 30 min of eccentric cycling at 100% concentric peak power. Instrumented-treadmill running at 10-18 km·h⁻¹ was tested pre, post, 24, 48, and 72 h. We quantified t_{push} , t_{brake} , and k_{vert} and analyzed each with mixed effects per speed (alpha 0.05) with post hoc. **Results:** Time effects appeared at 10-12 km·h⁻¹ ($p < 0.05$). At these speeds, t_{push} decreased post and approached baseline by 48-72 h (Fig. A), while t_{brake} showed no consistent change. For k_{vert} , 10 km·h⁻¹ showed lower values at 48-72 h than post (24 h intermediate) (Fig. B); 12 km·h⁻¹ showed no change. No effects were detected at 14-18 km·h⁻¹. **Conclusion:** After EIMD, low-speed running (10-12 km·h⁻¹) shows a transient deviation of the spring-mass rebound; t_{push} shortens and k_{vert} declines later at 10 km·h⁻¹ (Fig. A-B); by 72 h, restoration of t_{push} and k_{vert} indicates mechanical recovery of elastic optimization at low speeds.



LINKING GAINS AND LOSSES: REGIONAL ASSOCIATIONS BETWEEN MUSCLE HYPERTROPHY AND SUBCUTANEOUS FAT REDUCTION

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Introduction: Regional muscle hypertrophy is a well-established adaptation to resistance training when variables such as range of motion (ROM) are manipulated. However, it remains unclear whether such regional hypertrophy is accompanied by proportional reductions in localized subcutaneous fat. This study investigated whether muscle regions that hypertrophied more following a controlled resistance training protocol also exhibited greater regional fat loss. **Methods:** Nine untrained women completed 36 sessions of knee extension training over 12 weeks using a partial ROM (65°–30° of knee flexion). Pre- and post-training assessments included ultrasound measurements of rectus femoris cross-sectional area (CSA) and subcutaneous fat thickness at 40%, 50%, 60%, and 70% of femur length. Paired t-tests and ANOVA examined regional adaptations, and Pearson's correlations tested associations between hypertrophy and fat reduction. **Results:** Significant muscle hypertrophy occurred at 40% and 50% of femur length ($p \leq 0.021$), while fat thickness significantly decreased only at 40% and 60% ($p \leq 0.040$). ANOVA detected regional differences in hypertrophy ($p = 0.040$) but not in fat loss ($p = 0.262$). No associations were found between regional hypertrophy and regional fat reduction ($r = -0.19$ to 0.315 ; $p \geq 0.409$). **Conclusion:** Although muscle hypertrophy occurred in a region-specific manner, localized subcutaneous fat reduction did not follow the same pattern, and no relationship existed between hypertrophy magnitude and regional fat loss. These findings reinforce that fat reduction is predominantly systemic, not regionally influenced by localized muscular adaptations.

A PERIMETRIA MEDE O QUE PROMETE? COMPARAÇÃO ENTRE PERIMETRIA E ULTRASSONOGRAFIA NA DETECÇÃO DE INCHAÇO MUSCULAR AGUDO

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Introdução: A perimetria é um método acessível de avaliação física, contudo, sua sensibilidade para detectar o inchaço muscular agudo em comparação à precisão da ultrassonografia ainda é incerta. Portanto, o objetivo deste estudo foi analisar a confiabilidade da perimetria na avaliação do inchaço muscular do peitoral maior (PM) e tríceps braquial (TB) após o exercício supino reto, tendo a ultrassonografia como referência.

Métodos: Foram selecionados 20 homens (18-40 anos) com experiência prévia mínima de 6 meses ininterruptos em treinamento de força e sem histórico de lesões prévias em MMSS. Os participantes realizaram o exercício supino reto com barra livre ao longo de três sessões experimentais. Na primeira sessão, foi realizada a familiarização com o teste de 12 repetições máximas (12RM) no supino e a assinatura do Termo de Consentimento Livre Esclarecido (TCLE). Na segunda sessão, foi conduzido o teste de 12RM seguindo os mesmos procedimentos da familiarização. Na terceira sessão os participantes realizaram uma série com o número máximo de repetições com a intensidade absoluta (peso) encontrada no teste de 12RM. Antes e depois da realização dessa série, imagens de ultrassom modo B foram obtidas para mensuração da espessura muscular do peitoral maior (PM) e tríceps braquial (TB), bem como a circunferência do tórax e do braço calculada por perimetria. Antes e após a intervenção, mensurou-se a espessura muscular (ultrassom modo-B) e a circunferência (perimetria) do braço e tórax como indicativo de inchaço (via teste t pareado). Em seguida, a associação entre os inchaços foi verificada pelo Coeficiente de Correlação de Pearson.

Resultados: Para o PM, ambos os métodos detectaram um inchaço significativo: ultrassonografia ($3,12 \pm 0,68$ para $3,35 \pm 0,54$ cm; $p = 0,027$) e perimetria ($101,84 \pm 7,67$ para $103,28 \pm 7,69$ cm; $p < 0,001$). No TB, a ultrassonografia não indicou inchaço significativo ($p = 0,195$), divergindo da perimetria que apontou aumento ($33,92 \pm 3,78$ para $34,75 \pm 3,68$ cm; $p < 0,001$). Não houve correlação significativa entre os métodos para PM ($r = -0,354$) ou TB ($r = -0,113$).

Conclusão: De modo geral, a perimetria não se mostrou um método confiável para monitorar o inchaço muscular agudo nos grupos musculares avaliados, apresentando resultados divergentes e sem correlação linear com as alterações morfológicas agudas detectadas pela ultrassonografia.

IS MUSCLE SIZE A SYNONYM FOR POWER? EVIDENCE IN SNATCH PRACTITIONERS

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Introduction: This study aimed to examine the individual and combined associations of the cross-sectional areas (CSAs) of the vastus lateralis (VL) and rectus femoris (RF) with countermovement jump (CMJ) height in snatch (SN) practitioners. **Methods:** Nineteen recreational SN practitioners (10 women; 9 men), each with at least one year of uninterrupted training, participated in this cross-sectional study. Two experimental sessions were conducted: CMJ familiarization and, seven days later, panoramic ultrasound assessment of VL and RF CSAs followed by the official CMJ test. Ultrasound images were analyzed using HOROS software, with excellent reliability ($ICC \geq 0.95$). CMJ height was measured using the MyJump2 app. Simple linear regression analyses were performed to evaluate associations between muscle CSA variables and CMJ height, with significance set at $\alpha \leq 0.05$. **Results:** VL CSA showed a moderate and significant association with CMJ height ($R^2 = 0.41$; $p = 0.003$). The combined CSA of VL and RF also demonstrated a significant relationship with jump height ($R^2 = 0.44$; $p = 0.002$). RF CSA alone showed a positive but non-significant trend ($R^2 = 0.20$; $p = 0.054$). **Conclusion:** Both VL CSA and the combined VL + RF CSA were significantly associated with CMJ performance, indicating that muscle size—particularly of the VL—plays an important role in power expression among SN practitioners. RF CSA alone did not significantly predict CMJ height. These findings suggest that quadriceps morphology, especially VL hypertrophy, may serve as a practical indicator for monitoring and improving power-related performance in SN practitioners.

THE RELATIONSHIP BETWEEN PERCEIVED EFFORT, MUSCLE HYPERTROPHY, AND STRENGTH GAINS: DOES GREATER EFFORT LEAD TO GREATER RESULTS?

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Introduction: The present study aimed to analyze the association between ratings of perceived exertion (RPE) and the responses of muscle hypertrophy and maximal dynamic strength after a period of resistance training. **Methods:** For this purpose, ten healthy, untrained adult women underwent 12 weeks of knee extension training, performing 3–6 sets of seven repetitions at 60% of one-repetition maximum (1RM) with a three-minute rest between sets. At the end of the last set, the RPE (Borg scale: 6–20) was reported. Before and after the training period, the cross-sectional areas (CSA) of four regions of the vastus lateralis (VL), vastus intermedius (VI) and rectus femoris muscle (RF) were measured and summed, and maximal dynamic strength was assessed using the 1RM test. Spearman's correlation coefficient was used to verify the relationship between the percentage changes in individual and summed muscle CSA (CSA_summed), as well as 1RM performance, and the median RPE reported by participants across the training sessions. **Results:** The results indicated a weak and non-significant relationship between RPE and hypertrophy of VL ($\rho = -0.050$; $p = 0.892$), VI ($\rho = 0.186$; $p = 0.607$), and RF ($\rho = 0.235$; $p = 0.513$), as well as the CSA_summed ($\rho = 0.242$; $p = 0.501$). Likewise, no significant correlation was observed between RPE and maximal strength gains ($\rho = 0.025$; $p = 0.946$). **Conclusion:** Thus, the results suggest that there is no significant association between perceived exertion values and hypertrophy or strength outcomes under the training conditions used.

Efeitos de um programa comunitário de exercícios físicos na mobilidade funcional e flexibilidade de participantes idosos

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Introdução:

O Centro de Referência em Envelhecimento e Movimento (CREM), da Escola de Educação Física, Fisioterapia e Dança da UFRGS, é um programa comunitário que desenvolve modalidades de exercício físico voltados à promoção da saúde, autonomia e funcionalidade de pessoas idosas. As atividades incluem treinamento multicomponente, hidroginástica, musculação, equilíbrio e dança, sempre com foco na manutenção da funcionalidade e prevenção de quedas. A avaliação objetiva de mobilidade e flexibilidade é essencial para monitorar o impacto dessas práticas sobre indicadores motores fundamentais para a independência na vida diária.

Materiais e Métodos:

Foram incluídos 234 idosos com avaliações completas de pré e pós intervenção realizadas no ano de 2022 no CREM/UFRGS. A mobilidade funcional foi avaliada pelo teste Timed Up and Go (TUG). A flexibilidade foi avaliada por dois testes: sentado e alcançar para avaliar flexibilidade de membros inferiores (FlexMI) e alcançar atrás das costas para avaliar a flexibilidade de membros superiores (FlexMS). As comparações entre os momentos pré e pós-intervenção foram realizadas com o teste t pareado, adotando significância de 5%. Foram calculadas médias, diferenças médias, valores de p e tamanhos de efeito (d de Cohen). (falar mais dos testes?)

Resultados:

Foram observadas melhorias significativas nos três desfechos avaliados. O desempenho no TUG melhorou, com redução do tempo médio de 6,86 s (pré) para 6,47 s (pós), diferença de $-0,39$ s ($p < 0,001$; $d \approx 0,31$). A flexibilidade de membros inferiores (FlexMI) também apresentou ganho significativo, passando de $-3,12$ cm (pré) para $-1,92$ cm (pós), diferença média de $+1,20$ cm ($p = 0,027$; $d \approx 0,15$). Além disso, houve melhora significativa na flexibilidade de membros superiores (FlexMS), com valores médios de $-9,76$ cm (pré) e $-8,43$ cm (pós), representando um ganho de $+1,34$ cm ($p = 0,031$; $d \approx 0,14$). O padrão de melhora indica aumento da amplitude de movimento e maior alcance funcional após o período de intervenção.

Conclusão:

Os resultados demonstram que a participação no programa comunitário de exercícios físicos do CREM/UFRGS promoveu melhorias estatisticamente significativas na mobilidade funcional e na flexibilidade de membros superiores e inferiores. Os ganhos são coerentes com a literatura e clinicamente relevantes para a prevenção de quedas, a manutenção da autonomia e a execução de atividades de vida diária. Esses achados reforçam o valor de programas universitários comunitários como estratégia eficaz, acessível e baseada em evidências para promover envelhecimento ativo e funcionalidade em pessoas idosas.

Effects of Deep Water Running on the Functional Fitness of Older Adults: Analysis of Strength, Mobility, and Aerobic Capacity

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Introduction: Aging compromises muscular strength and cardiorespiratory capacity, influencing mobility and motor performance. Regular Deep Water Running practice can attenuate these effects, supporting the maintenance of functionality in older adults. **Objective:** To analyze the relationships between lower-limb strength, mobility, and aerobic capacity in older adults practicing aquatic jogging, as well as the effects of this practice on these components of functional fitness. **Methodology:** Twenty-six older adults practicing Deep Water Running in the CREM/UFRGS extension program participated in the study and were assessed before (pre) and after (post) a 9-month intervention period. The following tests were applied: the 30-Second Chair Stand Test (FMI30s) for lower-limb strength, the Timed Up and Go (TUG) test for functional mobility, and the 6-Minute Walk Test (6MWT) for cardiorespiratory capacity. Comparisons between moments were performed using Student's paired t-test, and associations between variables were analyzed using Pearson's correlation ($p \leq 0.05$), with the JASP software (v. 0.19.3). **Results:** The mean values ($\pm$ SD) in the pre and post assessments were respectively: FMI30s = 16.76 ± 5.25 and 16.83 ± 4.83 ; TUG = 7.26 ± 3.56 and 7.27 ± 4.12 s; 6MWT = 461.96 ± 126.07 and 518.53 ± 127.26 m. A significant improvement in aerobic capacity was observed after the intervention period (6MWT: $p = 0.006$), while strength (FMI30s: $p = 0.516$) and mobility (TUG: $p = 0.132$) did not show significant differences. In the correlational analysis, a strong negative association was found between TUG and 6MWT in the pre ($r = -0.692$; $p < 0.001$) and post ($r = -0.823$; $p < 0.001$), indicating that greater aerobic capacity is related to better mobility. Moderate negative correlations were also observed between TUG and FMI30s (pre: $r = -0.609$; $p = 0.001$; post: $r = -0.577$; $p = 0.012$), and a positive correlation between FMI30s and 6MWT in the post assessment ($r = 0.561$; $p = 0.019$). **Conclusion:** Deep Water Running improved aerobic capacity and maintained strength and functional mobility in older adults, highlighting the integration between strength, endurance, and motor control. These findings reinforce the role of Deep Water Running in preserving motor function and promoting healthy aging.

ABSTRACTS

Category/Topic 3: Sport or Motor Behavior and Rehabilitation



The Effect of Isometric Training with Biofeedback in Post-Stroke Individuals: Preliminary Data on Shoulder Function

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Introduction: Stroke is one of the leading causes of mortality and disability worldwide, resulting from the interruption of cerebral blood flow and subsequent neural damage. Among its sequelae, upper-limb motor deficits are particularly prominent, compromising daily activities and functional independence. In this context, the present study aims to examine the effects of upper-limb isometric training combined with biofeedback in individuals in the post-stroke period.

Materials and Methods: Eight individuals (N = 8) with a history of stroke, aged 60 to 80 years, participated in the study. Participants completed a 9-week, twice-weekly training protocol consisting of isometric training of the affected upper limb. The protocol required participants to perform an isometric shoulder extension at 45°, expressed as a percentage of their maximal voluntary isometric contraction (MVIC). Real-time biofeedback was provided via a computer screen displaying both the relative force produced and the target percentage. A Friedman ANOVA was conducted to compare MVIC values across the three assessment time points.

Results: The results did not reveal significant differences across the three time points [$X^2(2)$, $p = 0.06$].

Discussion/Conclusion: The preliminary findings of this study do not indicate significant effects of the intervention on MVIC. However, the p-value approached significance despite the small sample size of only eight participants. The study suggests that with a larger sample, the intervention might yield statistically meaningful differences.

Keywords: Biofeedback, Stroke, Gamification

The Application of Pupil Diameter Biofeedback May Enhance Motor Learning in Children and Adolescents with Autism Spectrum Disorder: A Pilot Study

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Introduction: During the acquisition of motor skills, the number of errors tends to decrease as the learner progresses through different stages. When practicing, individuals typically move from the cognitive stage to the autonomous stage. The autonomous stage, in particular, is marked by automaticity and reduced cognitive demand, where the performer exerts less conscious control over their actions and increasingly relies on implicit learning strategies. Within this framework, pupil diameter has been proposed as a sensitive marker of cognitive effort: greater dilation reflects higher effort, whereas smaller dilation reflects lower effort. Some studies have observed a gradual reduction in pupil size across motor practice, suggesting a link between cognitive load and pupillary responses. In individuals with Autism Spectrum Disorder (ASD), who often face pronounced difficulties in motor learning, it becomes essential to explore mechanisms that can facilitate skill acquisition. This study aims to investigate the effects of pupil dilation biofeedback on motor learning in children and adolescents with Autism Spectrum Disorder.

Materials and Methods: Three participants (two aged 7 and one aged 12), all right-handed and level 1 support, were recruited from LAIS and PRAIA at UFMG, Belo Horizonte. They first played a computer game using an eye-tracking interface that provided pupil dilation biofeedback, where smaller pupils helped an astronaut collect stars. In the biofeedback group, values closer to baseline relaxation were rewarded; in the placebo group, rewards were random. Afterward, participants performed a beanbag-throwing task at concentric targets, with practice on day 1 and retention/transfer tests after 24 hours. This pilot study aimed to assess the sensory adequacy of the headgear and the feasibility of applying the biofeedback protocol in children and adolescents with ASD.

Results and Discussion: The preliminary findings indicated no sensory discomfort and supported the feasibility of using the biofeedback protocol in a motor learning context. Although robust statistical analyses were not performed, there are indications that the protocol may contribute to enhancing motor learning in children with ASD.

Conclusion: The findings from this pilot study suggest that the proposed biofeedback protocol is promising, showing potential to help children and adolescents with ASD regulate a relaxed mental state, thereby supporting motor learning.

Acknowledgments: This work was supported by FAPEMIG – Call No. 016/2024/2024.

Optimizing Mental Effort During Motor Practice Through AI-Enhanced Pupil Biofeedback: A Motor Learning Study

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Introduction: Motor learning is accompanied by a progressive reduction in error and cognitive control demands, with mental effort playing a central role in the transition from controlled to automatic performance. Pupil dilation is a sensitive psychophysiological index of cognitive load and can be leveraged to probe and modulate ongoing mental states. This study investigated whether a human–computer interaction (HCI) system, combining a low-cost eye tracker and artificial intelligence, could optimize mental effort during motor practice by gating movement initiation to moments of reduced pupil-indexed effort.

Materials and Methods: Forty-two right-handed adults (18–35 years) performed a golf-putting task in three phases: pre-test, acquisition, and delayed learning tests (retention and transfer). In a two-stage design, a control group of 24 individuals practiced without feedback; these data were used to train a machine learning model that estimated a pupil-diameter threshold associated with improved performance. A biofeedback group then practiced with effort-contingent gating: an auditory cue signaled when pupil diameter remained below the individualized threshold during the planning period, prompting movement initiation. For between-group analyses, 18 biofeedback participants were matched to 18 controls. Primary and secondary outcomes included global error, pupil dilation, planning period duration, execution and preparation times, fixation duration, gaze shifts, and normalized jerk.

Results and Discussion: Global error decreased across practice in both groups, but the biofeedback group showed lower error during acquisition and in both retention and transfer tests. Biofeedback participants also exhibited greater reductions in pupil dilation, longer fixation duration, shorter execution and preparation times, and smoother strokes (lower normalized jerk) than controls. Within the biofeedback group, higher biofeedback mastery time and reach, as well as shorter initiation intervals, were strongly associated with lower error, indicating that better adherence to effort-contingent gating predicted superior performance.

Conclusion: Effort-contingent biofeedback on pupil dilation guided learners toward lower-effort mental states that supported more accurate, faster, and smoother putting, and yielded superior retention and transfer. These findings suggest that pupil-based HCI can modulate cognitive load during practice and promote movement automaticity, opening a promising avenue for integrating psychophysiological markers into motor learning interventions.

Acknowledgments: This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001

THE PRACTICE OF SHOTOKAN KARATE-DO ON REACTION TIME: A COMPARATIVE STUDY AMONG KARATE PRACTITIONERS, PRACTITIONERS OF OTHER MODALITIES AND NON-PRACTITIONERS

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Introduction:

In the context of Shotokan Karate-Do, reaction time is an essential physical skill directly linked to combat performance and self-defence, influenced by factors such as age, training duration, and rhythm. Given this, the present study aimed to investigate the influence of Shotokan Karate-Do practice on reaction time, comparing practitioners of this martial art with individuals from other sports modalities and non-practitioners.

Materials and Methods:

The research adopted a qualitative-quantitative, descriptive, and cross-sectional approach, with a sample composed of 28 Shotokan Karate-Do practitioners of different belt levels and 16 non-practitioners of various ages and training levels. The data collection instrument used was the Deary-Liewald CogniFit Test, applied to measure the participants' reaction time.

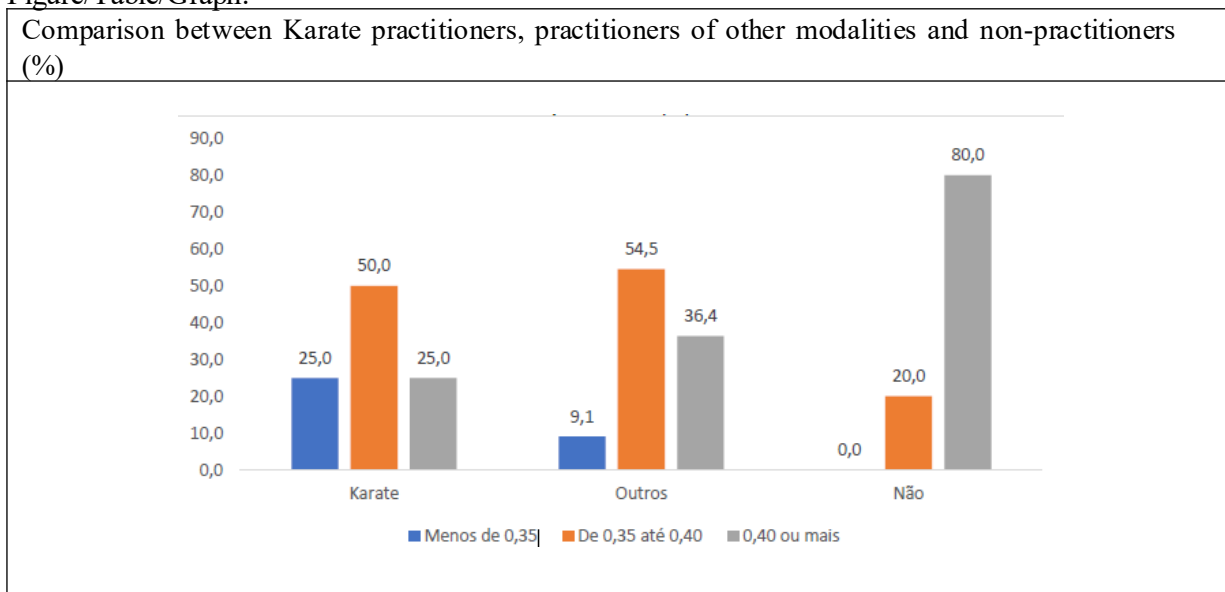
Results:

The results indicated that Karate practitioners showed shorter reaction times compared to non-practitioners and practitioners of other modalities, confirming the initial hypothesis. However, the group of teachers (black belts) showed inferior performance to the instructors (purple and brown belts), possibly because they dedicated less time to specific training and were more involved in teaching activities.

Discussion/Conclusion:

Concluded that the practice of Shotokan Karate-Do contributes significantly to the improvement of reaction time, reinforcing its importance as an instrument for physical and cognitive development. We recommended the realisation of longitudinal studies to monitor the evolution of this variable over the years of practice.

Figure/Table/Graph:



Effects of a multicomponent protocol with emphasis on muscle power training on bradykinesia in individuals with Parkinson's disease

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Introduction: Among the main symptoms of Parkinson's disease (PD) are bradykinesia and muscle weakness. Muscle power training may help reduce the effects of motor impairments, although evidence is still lacking for different disease subtypes. Therefore, this study aimed to investigate the effects of a multicomponent protocol emphasizing muscle power training on bradykinesia in individuals with the rigid-akinetic type of PD.

Materials and Methods: A Randomized Clinical Trial was conducted with 24 individuals diagnosed with PD (1-3 H&Y), who were randomized into two treatment groups: G1 - traditional multicomponent training, and G2 - multicomponent training with an emphasis on muscle power. Primary outcome motor function was assessed through the adapted Unified Parkinson's Disease Rating Scale (UPDRS III) (bradykinesia); followed by secondary outcomes muscle strength (1RM), muscle power of knee extensors (linear potentiometer) and body composition (DXA). Treatment sessions occurred twice a week for 12 weeks, lasting 60 minutes each.

Results: The multicomponent exercise protocol decreased UPDRS significantly both for G1 and G2, but there was no statistically significant difference between the two groups. Similarly, average power and 1RM strength increased significantly for both groups. There was no change in thigh fat-free mass.

Discussion/Conclusion: In conclusion, a multicomponent protocol emphasizing muscle power training and a traditional one were effective to reduce bradykinesia in individuals with rigid-akinetic type of PD, as well as strength and power, with similar results among them.

Figures:

A: Descriptive Table

Variable	T0		T1		T2		Follow Up	
	TRAD	POWER	TRAD	POWER	TRAD	POWER	TRAD	POWER
Mean Power (W)	37.58(14.99)	47.8(20.85)	66.37(28.63)	68.55(34.2)	69.03(31.98)	77.93(34.75)	73.44(35.93)	68.38(35.09)
LST (kg)	6.68(1.48)	6.46(0.91)	6.69(1.57)	7.01(1.05)	6.68(1.52)	6.94(1.27)	6.88(1.04)	6.92(1.21)
1RM (kg)	25.38(9.21)	25.55(11.51)	33.63(10.41)	29.18(11.54)	35.13(14.96)	33.27(13.96)		
UPDRS (au)	13.66(7.31)	13.18(6.50)	10.66(5.47)	10.36(6.93)	9.33(4.18)	10.36(6.60)		

B: ANOVA for repeated measures

ANOVA for repeated measures			
Outcome	Interaction	F	p
Mean Power	Time	9.876	0.001*
	Group	0.122	0.731
	Time * Group	0.606	0.52
1RM	Time	13.184	<.001*
	Group	0.15	0.704
	Time * Group	0.885	0.422
UPDRS	Time	3.887	0.030*
	Group	0.001	0.973
	Time * Group	0.184	0.833
MLG Thigh	Time	2.407	0.080
	Group	0.029	0.867
	Time * Group	1.682	0.184

* = p < 0.05.

SHOOTING PATTERNS ACROSS COMPETITION PHASES IN WOMEN'S 3X3 BASKETBALL: PARIS SUMMER OLYMPIC GAMES 2024

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Introduction: 3x3 basketball made its debut as an Olympic discipline at the Tokyo 2020 Games. Due to this recent inclusion, there is a clear need to investigate performance indicators to better understand the game demands. Time-motion analysis (TMA) is a tool already used in studies of this sport and proves useful in this process. Considering this, the aim of the present study was to compare shooting actions in 3x3 basketball matches from the Paris 2024 Olympic Games. **Materials and Methods:** This is a cross-sectional observational study that analyzed 34 3x3 basketball matches from the Paris 2024 Olympic Games, including both the group and knockout stages. Shooting attempts, successful shots, and shooting efficiency (%) for one and two points shots were considered. TMA was performed using the Longomatch software (Fluendo VAS SL, 0.20.8, Spain). For statistical analysis, means were compared using generalized estimating equations followed by Bonferroni post-hoc tests, adopting a significance level of $p < 0,05$. **Results:** Inferential analysis revealed a higher efficiency of one-point shots during the knockout stage ($p = 0,011$; $X^2 = 6,47$), while a higher number of successful two-point shots was observed during the group stage ($p = 0,012$; $X^2 = 6,26$). For the other variables, a similar action profile was found when comparing stages (Table 1). **Discussion/Conclusion:** The study observed a predominantly similar action profile, but with higher efficiency in one-point shots during the knockout stage and a greater number of successful two-point shots during the group stage. The similarity between stages may be explained by teams adopting the same scoring strategy across both phases (Ferioli et al., 2022). In conclusion, female 3x3 basketball athletes exhibit a similar shooting profile across different stages of the competition.

Table 1: Shooting profile in 3x3 female basketball matches, 2024 Olympic Games.

Number of Actions	Group Stage	Knockout Stage	p-value	X ²
	Mean $\pm$ SD	Mean $\pm$ SD		
1PT Successful Shots	2,13 $\pm$ 2,79	2,82 $\pm$ 4,66	0,053	3,753
1PT Shots Attempt	4,21 $\pm$ 4,40	3,99 $\pm$ 5,26	0,591	0,288
1PT Efficiency (%)*	47,07 $\pm$ 40,02	56,94 $\pm$ 41,55	0,011	6,468
2PTS Successful Shots	0,65 $\pm$ 1,36	0,26 $\pm$ 1,85	0,012	6,257
2PTS Shots Attempt	2,31 $\pm$ 4,35	1,93 $\pm$ 3,68	0,125	2,359
2PTS Efficiency (%)*	21,65 $\pm$ 42,83	34,57 $\pm$ 56,99	0,128	2,320

PT - Point; PTS - Points. * Data represented as percentages.

Comparison of gender performance and lower limb dominance in the Modified Star Excursion Balance Test

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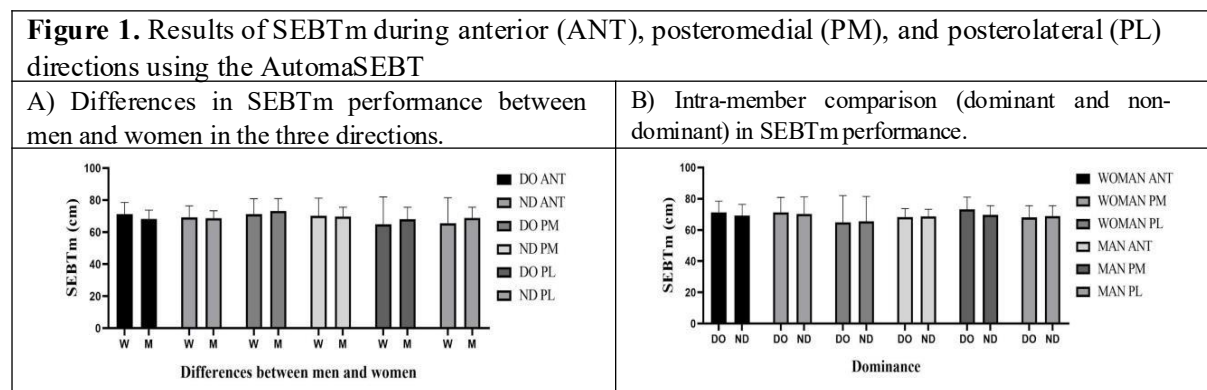
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Introduction: The Modified Star Excursion Balance Test (SEBTm) assesses dynamic postural control in different populations. Evidence on sex and limb dominance differences remains inconclusive. This study evaluated SEBTm performance in healthy adults using the AutomaSEBT, comparing dominant (DO) and non-dominant (ND) limbs and between sexes.

Materials and Methods: This cross-sectional study (no. 5.786.222) evaluated 30 healthy adults (18–40 years) without lower limb injuries in the past two years. The SEBTm was performed in the anterior (ANT), posteromedial (PM), and posterolateral (PL) directions using the AutomaSEBT, an automated device with an Arduino UNO-R3 and VL53L0X laser sensor that records reach distances via infrared pulses. Data normality was assessed with the Shapiro-Wilk test, and sex (male/female) and dominance (DO/ND) effects were compared with two-way MANOVA ($\alpha=0.05$).

Results: Our results showed no significant differences between sexes in the ANT, PM, and PL directions ($p>0.05$) (Figure A). Similarly, the comparison between DO and ND also showed no significant differences in any of the SEBTm directions analyzed ($p>0.05$) (Figure B). Figures 1A and 1B present results of the distances performance achieved in the three directions of the SEBTm. These findings indicate similar performance between men and women, as well as between limbs, suggesting functional symmetry in healthy individuals during the execution of the test.

Discussion/Conclusion: In summary, results presented no significant differences were found in SEBTm performance between sexes or between lower limbs. Our findings corroborate the literature, which reported balanced performance during. Furthermore, the use of AutomaSEBT facilitated the application of the SEBTm assessment, reducing observation errors and allowing the evaluator to focus on the criteria for correct execution of the respective test.



Muscle activation of the gluteus medius during the lateral step-down test in adolescents with and without patellofemoral pain

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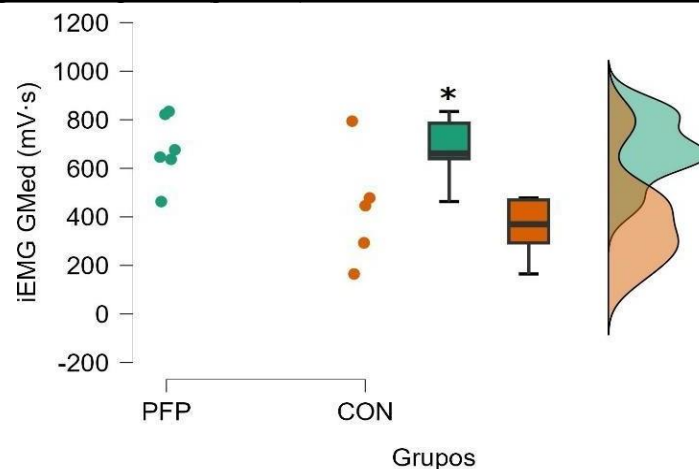
Introduction: Patellofemoral pain (PFP) is a common condition in adolescents, associated with overload on the patellofemoral joint. Alterations in gluteus medius (GMed) control can influence knee alignment and dynamic stability. Therefore, this study evaluated the muscle activation of the GMed during lateral step-downs in adolescents with and without PFP.

Materials and Methods: This cross-sectional study included 12 female adolescents, divided into a group with PFP (n=6) and a control group (n=6), with no anthropometric differences between them ($p>0.05$). The study was approved by the institutional ethics committee (CAAE: 26928514.6.0000.5346). Surface electrodes were placed on the GMed of the dominant lower limb according to SENIAM® recommendations. The lateral step-down test was performed on a step adjusted to produce 60° of knee flexion, in three sets of three repetitions. EMG signals were processed in MATLAB®, band-pass filtered (20–500 Hz), low-pass filtered (15 Hz, 4th order Butterworth), and normalized by peak amplitude. Comparisons between groups were performed using the independent t-test ($\alpha=0.05$).

Results: The PFP group showed greater GMed muscle activation (39.5% higher compared to the control group) and a large effect size ($d=1.466$), as illustrated in Figure 1.

Discussion/Conclusion: Adolescents with PFP presented greater activation of the GMed compared to the control group, possibly as a compensatory strategy to stabilize the hip and knee during movement. This response may represent an adaptive mechanism to joint instability and strength deficits, but it may also indicate greater neuromuscular effort and risk of fatigue.

Figure 1. Gluteus medius muscle activation (iEMG GMed) in the patellofemoral pain (PFP) and control (CON) groups during lateral step down ($p=0.029$).



Funções Executivas em Adolescentes: Influência do Treinamento de Beach Tennis

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Introdução: O desenvolvimento das funções executivas (controle inibitório, memória de trabalho e flexibilidade cognitiva) tem sido associado à prática esportiva. O objetivo deste estudo foi investigar a influência de um programa de Beach Tennis nas Funções Executivas de adolescentes.

Materiais e métodos: O programa de BT foi implementado ao longo de 3 meses, com práticas regulares específicas do esporte 2x por semana por 60 minutos cada. Os participantes (N = 22; 11-14 anos) foram avaliados individualmente em três tarefas computadorizadas (setas, seis formas abstratas e figuras), antes e após o treinamento. O percentil do Score Z do número de respostas corretas nas tarefas e uma ANOVA de medidas repetidas foram utilizados.

Resultados: Na tarefa de alta demanda de controle inibitório (setas), uma interação não significativa entre grupo e tempo foi detectada, $\Lambda = 0.98$, $F(1,20) = 0.40$, $p = 0.534$, $\eta^2 = 0.20.4$; o fator grupo de idade também não foi significativo, $F(1,20) = 1.15$, $p = 0.699$, $\eta^2 = 0.008$; entretanto, o fator tempo evidenciou mudanças significativas e positivas do pré- para o pós-intervenção, $F(1,20) = 11.39$, $p = 0.003$, $\eta^2 = 0.363$. Nos demais testes as interações grupo e tempo não foram significativas, na tarefa de alta demanda de memória de trabalho (seis formas: $\Lambda = 0.92$, $F(1,20) = 1.61$, $p = 0.218$, $\eta^2 = 0.075$) e na tarefa de média demanda de controle inibitório e baixa demanda de memória de trabalho (figuras: $\Lambda = 0.99$, $F(1,20) = 0.22$, $p = 0.882$, $\eta^2 = 0.001$); para essas tarefas o fator grupo de idade (seis formas: $F(1,20) = 0.02$, $p = 0.876$, $\eta^2 = 0.001$ e figuras: $F(1,20) = 0.61$, $p = 0.441$, $\eta^2 = 0.030$) e o fator tempo (seis formas: $F(1,20) = 0.09$, $p = 0.927$, $\eta^2 = 0.000$ e figuras: $F(1,20) = 1.26$, $p = 0.274$, $\eta^2 = 0.059$) também não foram significativos.

Discussão/conclusão: Na tarefa “setas” a mudança significativa de todos os participantes do pré- para o pós-intervenção evidencia uma possível influência positiva da intervenção. Apesar disso, os resultados devem ser interpretados com cautela em decorrência das limitações do estudo – falta de grupo controle e da falta de controle das variáveis intervenientes que também podem contribuir para a melhora de controle inibitório.

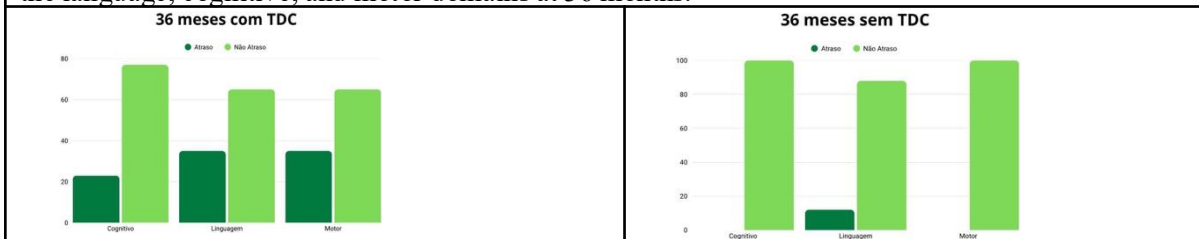
NEURODEVELOPMENTAL TRAJECTORIES IN PRETERM CHILDREN WITH BPD: A COMPARATIVE ANALYSIS BETWEEN GROUPS WITH AND WITHOUT DCD

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Introduction: Premature children often face health complications due to physiological immaturity. Bronchopulmonary Dysplasia (BPD), a chronic lung disease requiring prolonged oxygen support, increases vulnerability to adverse clinical outcomes and neurodevelopmental issues. Developmental Coordination Disorder (DCD) involves persistent motor coordination difficulties. This study aims to compare children with BPD, with and without DCD, regarding neurodevelopmental delays during the first 36 months of life. **Materials and Methods:** A longitudinal cohort study was conducted at the Hospital de Clínicas de Porto Alegre, following preterm infants enrolled in a program that monitors all children born preterm with low birth weight (<1500 g). A total of 43 children with Bronchopulmonary Dysplasia were evaluated, 26 with DCD and 17 without the disorder. The Bayley Scales of Infant and Toddler Development–III were used to assess the cognitive, motor, and language domains at 6, 12, 24, and 36 months. Children with DCD were diagnosed using the MABC-2 battery. **Results:** The Chi-square results show that, for children with BPD and DCD, the prevalence of delay is significantly higher at 36 months in the cognitive ($p = 0.006$) and motor ($p = 0.003$) domains. At earlier ages (6 and 12 months), the performance of children with BPD with and without DCD was similar. Children with BPD show higher prevalence rates of DCD (REF), similar to the present study. These children also show a higher prevalence of delays. **Discussion/Conclusion:** In summary, the study reinforces that DCD, when present in preterm children with BPD, indicates a trajectory of greater neurodevelopmental risk, and its early identification is essential for timely and more effective interventions.

Legend: Result tables of delays and non-delays in children with and without DCD and with BPD in the language, cognitive, and motor domains at 36 months.



Efeitos de um Programa de Intervenção Esportiva no Desenvolvimento das Habilidades Motoras Fundamentais em Crianças

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Introdução: O presente estudo teve como objetivo verificar o impacto de um Programa de Intervenção com base no ensino do Tênis, com uma abordagem de Contexto Motivacional para a Maestria, sobre o desenvolvimento das habilidades motoras fundamentais de crianças.

Materiais e Métodos: A amostra desta pesquisa foi composta por 26 crianças, com idades de 5 a 7 anos, participantes de um Projeto de Iniciação ao Tênis, desenvolvido em 16 semanas. Para as avaliações das crianças foi utilizado o *Test of Gross Motor Development-2* (TGMD-2) (ULRICH, 2000).

Resultados: Os resultados indicaram melhoras significativas no desempenho motor geral das crianças após o período de intervenção, passando de M=75,54 (pré-teste) para M=82,81 (pós-teste) (Tabela 1). No subteste de locomoção a média foi de M=30,58 para M=34,77 (p=0,000) enquanto no subteste de controle de objeto a média passou de M=28,77 para M=31,85 (p=0,016). Quanto à análise das categorias descritivas do teste (Tabela 2), a maioria das crianças encontrava-se na categoria “Pobre” (38,5%) e “Muito Pobre” (23,1%). Após o Programa, observou-se aumento expressivo da categoria “Média” (de 19,2% para 34,6%) e redução da categoria “Muito Pobre” para 3,8%.

Discussão/Conclusão: Conclui-se que, programas esportivos sistematizados e orientados por uma abordagem motivacional são eficazes para promover melhorias significativas no desenvolvimento das habilidades motoras fundamentais em crianças (BERLEZE, VALENTINI, 2021, 2022; VALENTINI, 2002; PÍFFERO, 2007).

Tabela 1: Análise Descritivas - Desempenho Motor Geral

TGMD2	Pré			Pós		
	Média	Min	Máx	Média	Min	Máx
Geral	75,54	52,00	91,00	82,81	67,00	100,00
Manipulativas	28,77	14,00	40,00	31,85	26,00	41,00
Locomotoras	30,58	13,00	38,00	34,77	27,00	43,00

Tabela 2: Categorias Descritivas do Desempenho Motor

Categorias Descritivas	Geral			
	Pré		Pós	
	(f)	%	(f)	%
Muito Superior	-	-	-	-
Superior	-	-	-	-
Acima da Média	-	-	-	-
Média	05	19,2	09	34,6
Abaixo da Média	05	19,2	05	19,2
Pobre	10	38,5	11	42,3
Muito Pobre	06	23,1	01	3,8
Total	26	100	26	100

Critical Speed

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Introduction: Rowing is a 2000-meter water sport that requires appropriate training load management strategies. On this sense, critical speed (CS) emerges as a practical and low-cost alternative for training prescription. Objective: To describe heart rate (HR) responses during a training session based on CS.

Materials and Methods: Ten male rowers (16.2 ± 1.23 years) participated in the study. Dependent variables were derived from HR and Rating of Perceived Exertion (RPE). Maturation was assessed using peak height velocity (PHV). CS was estimated from the slope of the distance–time regression line. HR was recorded using Polar H10 sensors, and RPE was measured with the Borg scale (6–20). The training pace was set at 100% of CS, and the session continued until voluntary exhaustion or was terminated after two times outside the target pace. Data were analyzed using descriptive statistics (mean and standard deviation).

Results: PHV indicated a post-pubertal stage. The mean exercise duration was 2.24 ± 2.51 min, with a stroke rate of 29.52 ± 7.47 strokes/min, power output of 316.65 ± 88.11 W, and a pace of $1:36.0 \pm 0:03.0$ min/500 m. Peak HR reached 189.60 ± 39.86 bpm ($96.91 \pm 3.87\%$ of HRmax), while the mean HR was 175.30 ± 5.65 bpm ($89.90 \pm 2.89\%$ of HRmax).

Discussion/Conclusion: It is concluded that CS-based training prescription led athletes to sustain high effort levels, demonstrating the applicability of CS as a valid parameter for intensity control in young rowers.

Injury Risk in Taekwondo: An Assessment Based on the Landing Error Scoring System in Athletes from Pelotas/RS

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Introduction: Taekwondo is a high-intensity sport that involves explosive movements, such as jumping kicks, which increase the risk of lower-limb injuries. In this context, the Landing Error Scoring System (LESS) stands out as a practical and accessible tool to assess landing patterns and identify injury risk. Thus, the aim of this study was to analyze lower-limb injury risk in Taekwondo athletes from Pelotas/RS using the LESS test. **Materials and Methods:** This observational and descriptive study included 20 Taekwondo athletes (19.67 ± 0.58 years), all physically active and eligible for testing. The LESS test was applied, consisting of 17 criteria evaluated using two cameras positioned in the frontal and sagittal planes. Each athlete performed five jump trials from a 30 cm box, and the videos were analyzed in slow motion (Kinovea 2023.1). Scoring assigned 1 point for each recorded error. Statistical analysis was conducted using STATA 17 and included the calculation of Cronbach's alpha to assess data reliability, showing internal consistency of $\alpha \geq 0.721$, classified as substantial according to Landis. **Results:** The LESS scores showed substantial variation among the athletes, ranging from 1.8 to 11 errors. The most frequent errors were insufficient hip and knee flexion, landing stiffness, and knee valgus, highlighting marked differences in the quality of movement patterns. **Discussion/Conclusion:** The results indicate that several athletes exhibit landing patterns associated with a higher risk of injury, particularly at the knee, underscoring the importance of ongoing monitoring through functional assessments and preventive training. The findings also show that the LESS proved to be an effective tool for identifying these risks and guiding prevention strategies in Taekwondo.

Figure/Table/Graph:

Table 1. Distribution of age, Taekwondo belt level, and LESS score among the 12 athletes analyzed

	A	B	C	D	E	F	G	H	I	J	K	L
AGE	16	17	16	25	25	14	16	14	14	14	19	16
BELT	Black	Light Red	Red	Light Red	Dark Blue	Dark Blue	Light Green	Orange	Gray	Gray	Gray	Light Blue
LESS SCORE	8,2	7,4	8,4	1,8	4,6	8,0	1,8	9,4	9,6	11,0	4,4	9,4

Functional and conventional hamstrings-to-quadriceps ratio: inter limb asymmetries in sports

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Introduction: Hamstring-to-quadriceps (H:Q) ratio represents the most common parameter to estimate muscle strength balance. Each modality has specific characteristics that can be affect the asymmetries. Thus, the purpose of this study is comparing functional and conventional H:Q ratio of different modalities. **Materials and Methods:** A total of 103 athletes were included from different sports modalities. Anthropometric data was collected, and knee extension/flexion torque were collected using isokinetic dynamometer. Subjects were positioned with the hip at 85° of flexion and knee joint were aligned with dynamometer rotation axis. Warm-up consisted of 20 submaximal concentric repetitions at 120°/s. Then, five maximum concentric repetitions of flexion/extension followed by five excentric knee flexion at 60°/s. The torque values were used to calculate the conventional (0.6 normal ratio) and functional (1.0 normal ratio) H:Q ratio. For normality and homogeneity of the data were utilized the Kolmogorov-Smirnov test. A Man-Whitney U test was used to compare functional and conventional ratio in dominant and non-dominant limb between athletes. All the tests were analyzed with software SPSS® (22.0), with significance level of $\alpha=0.05$. **Results:** Means of conventional H:Q ratio were 0.57 ± 0.10 and 0.53 ± 0.10 in dominant limb, respectively. Furthermore, functional H:Q ratio 0.91 ± 0.15 and 0.89 ± 0.20 in non-dominant limb, respectively. Likewise, Man-Whitney U test shown that there was dominant limb is larger than conventional and functional H:Q ratio to non-dominant ($p < 0.05$). **Discussion/Conclusion:** In conclusion, the results demonstrated dominant conventional and functional H:Q ratio compared non-dominant. Furthermore, both limbs presented imbalance H:Q ratio compared reference values. These results suggested hamstrings weakness in the athletes. Moreover, these findings presence of inter-limb strength asymmetries, which may have implications for performance and injury prevention.

Table 1. Comparison between dominant and non-dominant limbs in conventional and functional hamstring-to-quadriceps (H:Q) ratio in sports.

Variable	Dominant Limb	Non-dominant Limb	p-value
Conventional H:Q Ratio	0.57 ± 0.10	0.53 ± 0.10	$< 0.05^*$
Functional H:Q Ratio	0.91 ± 0.15	0.89 ± 0.20	$< 0.05^*$

Significant Influences of Prematurity with Different Gestational Ages and DCD Diagnosis on the Prevalence of Motor Development Delay

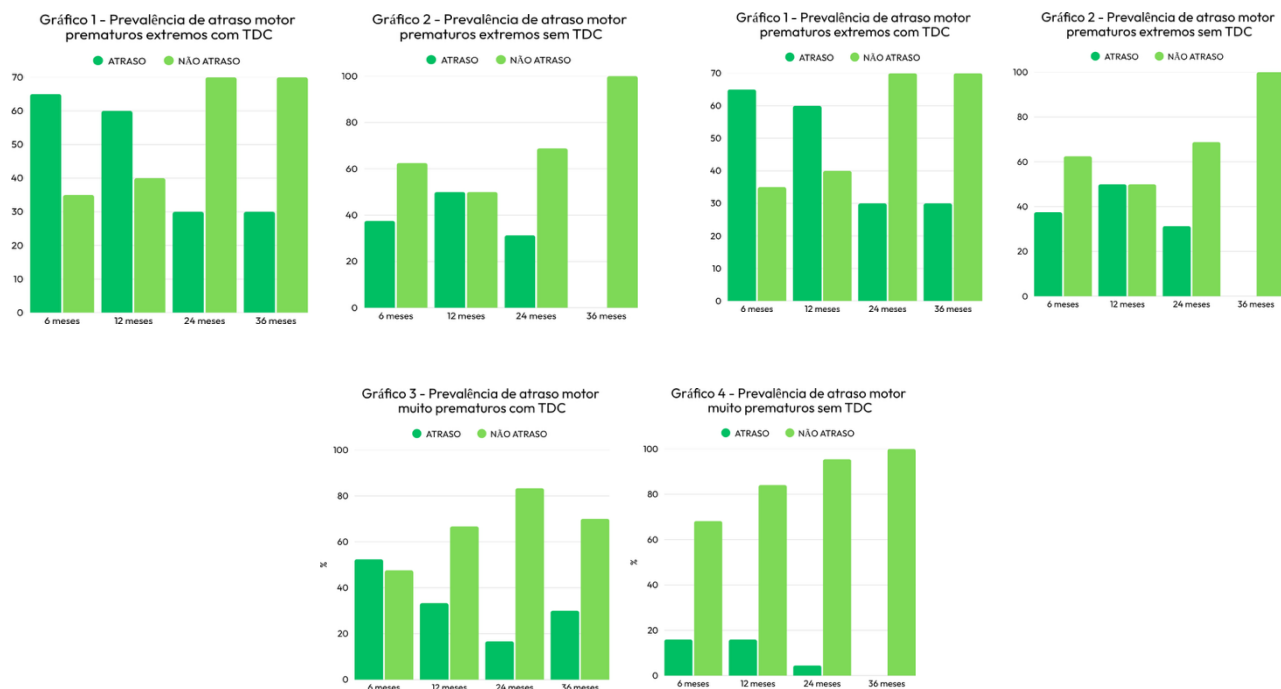
WEIRICH, Caroline¹; PANCERI, Carolina²; PESSI, Giovana¹; MAIA, Vitória³; SILVEIRA, Rita C⁴, VALENTINI, Nádia Cristina⁵.

Introduction: Preterm newborns have higher rates of neurodevelopmental impairment due to physiological immaturity and clinical manifestations proportional to the infant's gestational age. Developmental Coordination Disorder (DCD) can be recognized in children with movement difficulties, which significantly impacts their activities of daily living, school performance, and social participation. This study compares the significant influences of preterm children with different gestational ages diagnosed with DCD throughout the first 36 months of life.

Materials and Methods: 150 preterm children (75 with DCD and 75 without DCD) categorized by gestational age were evaluated using the BSITD-III in the motor, cognitive, and language domains at 6, 12, 24, and 36 months. Preterm children with DCD had their diagnosis confirmed at 4 years of age using the M-ABC-2 battery.

Results: The results show that in the extremely preterm group, children without DCD exhibit a lower prevalence of motor delay and a decline in motor delay indices over the months. Extremely preterm infants with DCD already present a persistent pattern of motor delay starting at 6 months, suggesting a more compromised and continuous motor trajectory due to physiological vulnerabilities. In the very preterm group with DCD, there is an increase in delay indices in the first months and stabilization at 24 and 36 months. However, in the group without DCD, the prevalence indices of motor delay are evidently lower compared to those with DCD. Finally, in the moderately preterm group, the motor prevalence index is highest at 36 months. Conversely, in the group without DCD, the children showed a very favorable motor trajectory in the later months.

Discussion/Conclusion: In summary, the results demonstrate that the highest indices of motor delay occur in the first months of life, especially among the most immature preterm infants. This pattern reinforces the need for early motor interventions, capable of mitigating the effects of physiological complications of prematurity and promoting more appropriate motor trajectories, both in children who progress to DCD and those who show recovery throughout childhood.



Controle inibitório e habilidades motoras: associações em crianças praticantes e não praticantes de atividades motoras

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RESUMO:

Introdução: As transformações motoras ao longo da infância ocorrem paralelamente ao desenvolvimento das funções executivas (FE) — controle inibitório, memória de trabalho e flexibilidade cognitiva. A relação entre habilidades motoras e funções executivas costuma ser atribuída à participação de áreas cerebrais que atuam em ambos os domínios. As redes neurais que envolvem o córtex frontal, parietal e motor não sustentam apenas os processos executivos, mas também desempenham papel central na realização de tarefas motoras. No cotidiano, as crianças precisam recorrer ao pensamento criativo para inventar novas brincadeiras, tanto no contexto esportivo quanto em outros domínios do desenvolvimento, e exercer controle inibitório para resistir a impulsos e agir de maneira mais ponderada. O objetivo deste estudo foi analisar a associação entre desempenho motor (habilidades locomotoras e com bola) e o controle inibitório de crianças praticantes e não praticantes de atividades motoras extracurriculares. **Material e métodos:** Participaram da pesquisa 247 crianças (4 a 10 anos). Para a avaliação do desempenho motor, foi utilizado o teste TGMD-3 (ULRICH, 2017), enquanto o controle inibitório foi mensurado pela bateria proposta por Davidson et al., (2006). **Resultados:** Correlações significativas e positivas foram observadas para (1) crianças praticantes de atividades motoras entre o controle inibitório (alta demanda¹) e habilidades com bola ($r=0,375$; $p\leq 0,001$) e entre controle inibitório (alta demanda²) e habilidades locomotoras ($r= 0,235$; $p= 0,008$) e com bola ($r= 0,442$; $p\leq 0,001$) e (2) para crianças não praticantes de atividades motoras entre o controle inibitório (alta demanda^{1 e 2}) e as habilidades com bola (respectivamente $r=0,269$; $p= 0,003$; $r= 0,307$; $p=0,001$). **Conclusões:** Os resultados indicam que o controle inibitório se relaciona ao desempenho motor tanto em crianças praticantes quanto em não praticantes de atividades motoras extracurriculares ao longo da infância. Esses resultados ressaltam a importância de considerar simultaneamente aspectos motores e cognitivos no acompanhamento do desenvolvimento infantil.

Shoulder Strength and Muscle Thickness Following Supraspinatus Tendon Repair: Partial Results of a Comparative Study with Healthy Controls

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Introduction: Supraspinatus tendon ruptures lead to short-term structural and functional deficits. However, the long-term effects of surgical repair (i.e., one-year after surgery) and physical exercise on the recovery of muscle thickness and shoulder abductor strength remain unclear. Furthermore, it is not well established whether the contralateral shoulder, often used as a reference of normality, can be considered healthy. Therefore, the aim of this study was to evaluate and compare shoulder abduction torque and the thickness of the supraspinatus and acromial deltoid muscles between individuals who underwent supraspinatus tendon repair and healthy controls one year after surgery. We also aimed to verify whether the uninjured side can serve as a reference for a healthy side.

Materials and Methods: Twenty-three physically active participants over 60 years of age, of both sexes, were included and divided into two groups: the supraspinatus tendon repair group (n = 12; mean age = 69.2 years; women = 9; men = 3) and the healthy control group (n = 11; mean age = 68.3 years; women = 8; men = 3). Participants were matched for age and sex. Shoulder abduction isometric torque was assessed using an isokinetic dynamometer, while supraspinatus and acromial deltoid muscle thickness were measured by ultrasound. Both sides were assessed in all participants. The significance level was set at $\alpha \leq 0.05$.

Results: The groups were similar for sex, age, height, body mass, physical activity level, and upper limb dominance ($p > 0.05$). No significant differences were observed between groups or sides for shoulder abduction isometric torque. Similarly, there were no differences in supraspinatus or acromial deltoid muscle thickness between groups or sides ($p > 0.05$).

Discussion/Conclusion: Individuals who underwent supraspinatus tendon repair demonstrated shoulder abduction torque and supraspinatus and acromial deltoid muscle thickness values similar to those of healthy controls. These findings suggest that surgical repair is effective for the long-term structural and functional recovery of the shoulder. Moreover, the fact that participants were physically active may have contributed to the similarity between groups, reinforcing the important role of physical exercise in restoring and maintaining shoulder function. Finally, the results indicate that the contralateral shoulder may serve as a valid reference of normality in studies involving physically active older adults.

Multicomponent Exercise Protocol with Emphasis on Muscle Power Training for the Treatment of Bradykinesia in Individuals with Parkinson's Disease

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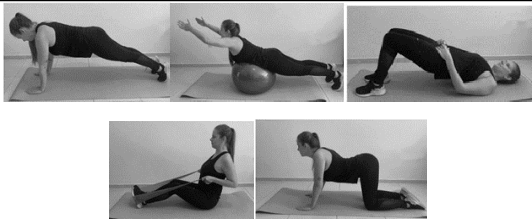
Introduction: Individuals with Parkinson's disease of the rigid-akinetic subtype present symptoms such as bradykinesia, muscle weakness, and reduced power. Most physiotherapy treatments focus on balance, mobility, strength, and gait, but do not include multicomponent training with emphasis on muscle power, as seen in other populations, not describe detailed protocols. This study aims to investigate the effect of a multicomponent protocol emphasizing muscle power training on bradykinesia in this population.

Materials and Methods: This study presents a detailed protocol applied to two groups: G1 – traditional multicomponent training (specific exercises, 2 sets of 10 repetitions) and G2 – multicomponent training with emphasis on muscle power (progressive load from 35% to 60% of 1RM), twice a week for 12 weeks. Bradykinesia (UPDRS III), power, flexibility, muscle rigidity, and functional capacity will be assessed. Each of the 20 exercises has three difficulty levels to adapt to participants' limitations.

Expected Results: It is expected that this protocol will help both groups reduce classic symptoms such as bradykinesia and muscle weakness, and prove to be safe and feasible for this population.

Discussion/Conclusion: Working with individuals with Parkinson's disease is challenging, as it is a chronic, progressive, and non-linear condition involving a highly heterogeneous population.

Figure/Table/Graph: *Photos of the researchers themselves*

Legend: Example of Exercises from the Multicomponent Power Training Protocol	
A Flexion, Row, Bridge, Trunk Extension.	B Leg extension machine, Stair, Step.
	

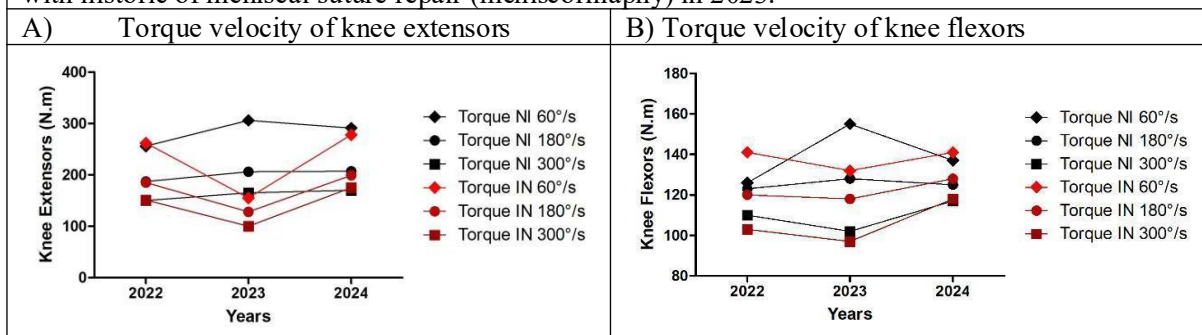
Effects of meniscal suture on knee extension and flexion torque in a futsal athlete: a case study.

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Introduction: Meniscal injuries are common in sports that involve abrupt changes in direction and may lead to reduced strength and neuromuscular control, particularly of the quadriceps muscle. In young athletes, meniscal repair surgery, aimed at preserving the meniscus, has been recommended due to the long-term implications of meniscal damage. Therefore, aimed to evaluate the knee flexor and extensor torque following meniscal suture repair (meniscorrhaphy) in a futsal athlete. **Materials and Methods:** A male futsal athlete, aged 18 years, weighing 86.7 kg and measuring 188.1 cm (2022, pre-injury), was evaluated with an isokinetic dynamometer (Biodex System 4, EUA). A protocol of five repetitions of knee flexion and extension at 60°, 180° e 300°/s, with a two-minutes rest interval between tests were performed. Strength assessments were conducted at three time points: pre-injury (May 2022), five months post-surgery (June 2023) and 14 months post-surgery (March 2024). Peak torque values were compared between the non-injured (NI) and injured (IN) limbs for both knee flexors and extensors. **Results:** The athlete presented at pre-injury a symmetric torque between sides (−2% to 1% for extensors and −12% to 6% for knee flexors). However, at five months post-meniscal suture, considerable torque asymmetry appeared in IN limb (extensors: 38%–49% deficit; flexors: 5%–15% deficit). Fourteen months after surgery repair (March 2024), the athlete's knee extensor torque returned to symmetry (Figure 1). **Discussion/Conclusion:** In summary, the findings revealed a substantial loss of knee extensor torque five months after meniscorrhaphy surgery repair, followed by recovery of these functional torque production 14 months after meniscorrhaphy surgery repair. These results indicating neuromuscular improvement and a return to baseline torque levels, and reinforces the importance of functional rehabilitation in restoring muscular torque after injury.

Figure 1. Muscle knee extensors and flexors torque over three years (2022-2024) of futsal player with historic of meniscal suture repair (meniscorrhaphy) in 2023.



THANK YOU!!!



OBRIGADO!!!